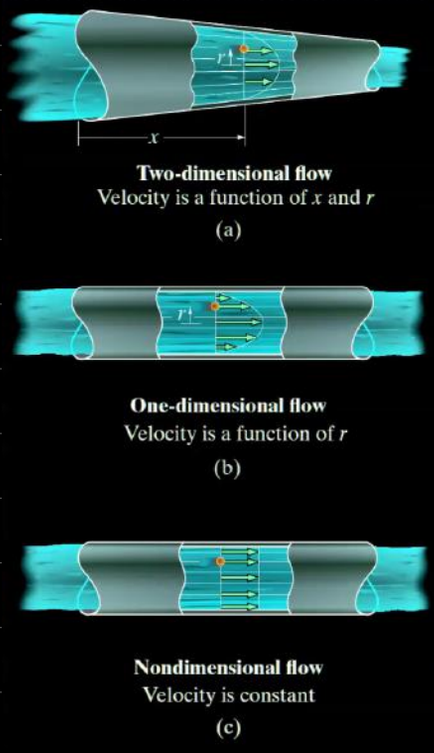
Description Of Fluid Flow:

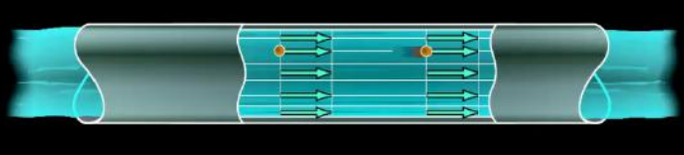
In a typical fluid flow, very huge number of particles are involved and accordingly describing fluid flow in terms of motion of individual particles will be cumbersome. For most applications we are not interested in exact motion of individual particles, so we visualise the flow to be a continuum of moving and deforming elemental volumes. Accordingly we describe the fluid flows in terms of flow field variables such as velocity $\vec{V}(x,y,z,t)$, Pressure : $P(x,y,z,t)$, density : $\rho(x,y,z,t)$ and acceleration : $\vec{a}(x,y,z,t)$ etc. A field variable associated with flow may either be a scalar (such as Pressure or density) or a vector (such as velocity and acceleration).

Classification Of Fluid Flows Based On Dimensions:

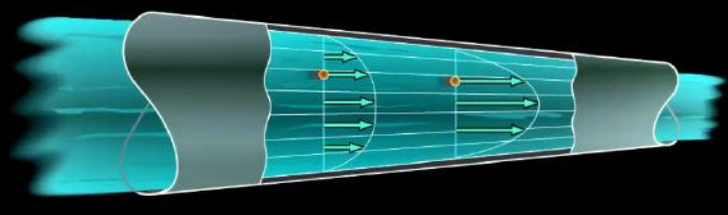
Based on number of variables required to describe velocity at any point in the fluid, the flows can be classified as Two dimensional, one dimensional or non dimensional (see figure for understanding the difference). There can also be three dimensional flows where the velocity depends on all three direction (x, y & z). However for JEE purposes we will be mostly dealing with flows being uniform over a given cross section of pipe (except in viscosity where radial velocity variation will also be considered).



Steady Flow:



Steady uniform flow
An ideal fluid maintains the same velocity at all times (steady) and at every point (uniform)

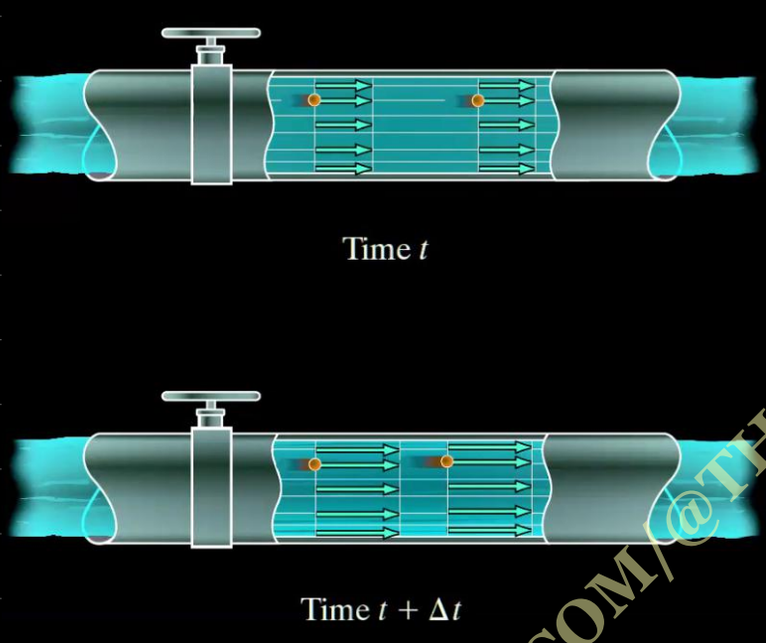


Steady nonuniform flow
The velocity remains constant over time (steady), but it is different from one location to the next (nonuniform)

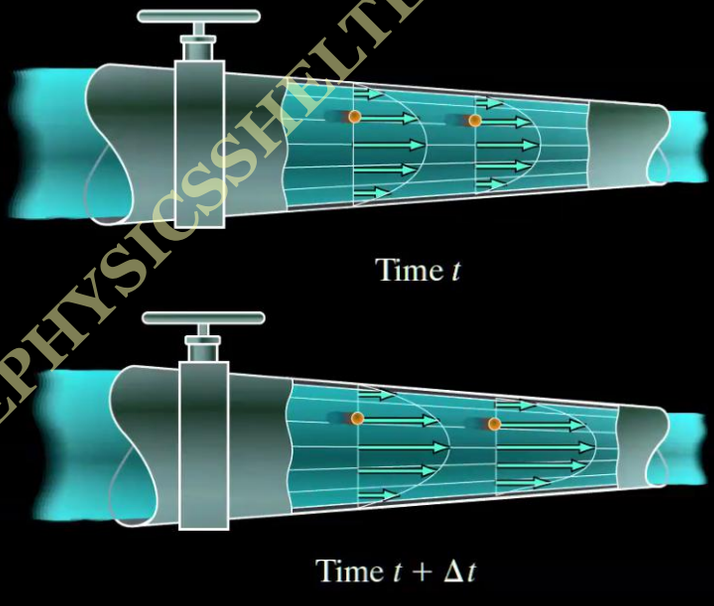
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If the velocity at any point in a tube does not change with time, the flow is said to be steady. Steady flow can be further classified as uniform or non-uniform. The velocity is constant over the cross section of the tube, the flow is steady, if it varies with choice of point at a given cross section we call is non-uniform flow. In our study for JEE we will be mostly dealing with steady flows.

Unsteady Flow:



Unsteady uniform flow
The valve is slowly opened and so at any instant the velocity of an ideal fluid is the same at all points (uniform), but it changes over time (unsteady)



Unsteady nonuniform flow
The valve is slowly opened, and because of the changing cross section of the pipe, the velocity will be different at each point (nonuniform) and at each time (unsteady)

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If the velocity at any point in a tube varies with time, the flow is said to be unsteady. Unsteady flows can be further classified as uniform and non-uniform unsteady flows. (see figure above).

Classification Of Fluid Flow Based On Frictional Effects:

Laminar Flow: When fluid particles move in form of neat layers (such as concentric layers in cylinders or laminae along plane surfaces) we call the flow as laminar. In this type of flow, each particle remains confined to its layers.

Turbulent Flow: At high velocities (or low viscosities), fluid particles tend to follow erratic paths thus leading to a mixing of fluid. This type of flow is called turbulent flow.

Transition Flow: A region in which both turbulent and laminar flow coexist is called transition flow.



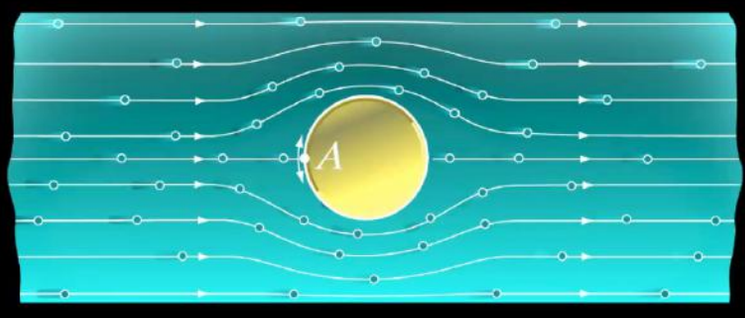
Laminar and turbulent flow from the smoke of an extinguished candle

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Graphical Description Of Fluid Flow:

Streamlines:


Velocity field around a cylinder

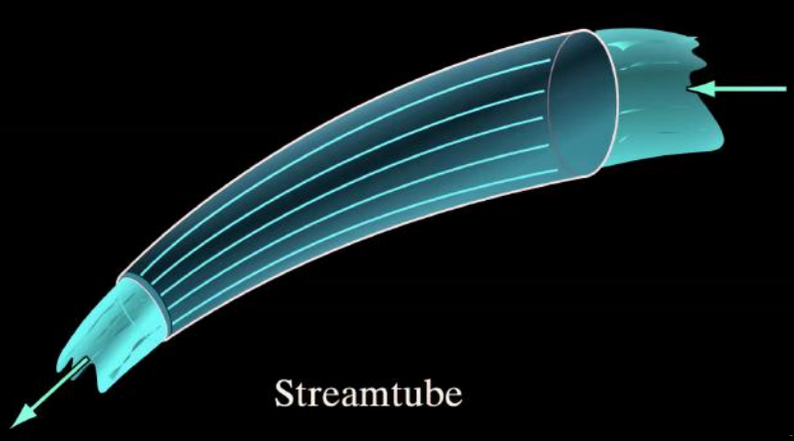


Streamlines for ideal fluid flow around a cylinder

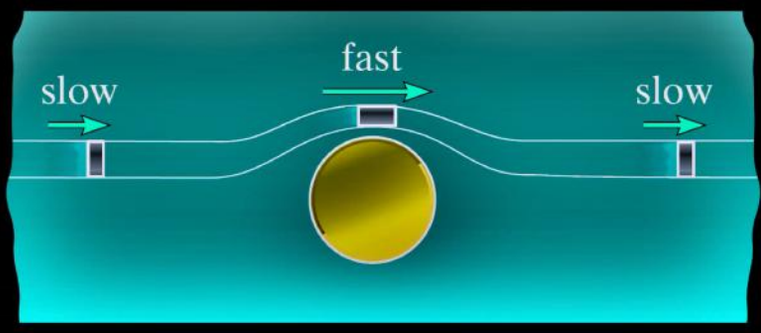
Streamlines are drawn in a flow field to visualise the velocity variation at various points. The density of streamlines is proportional to magnitude of local velocity while their tangent gives direction of velocity at that location. Streamlines never intersect each other. Whenever streamlines have to take a U turn (due to some obstacle), the local velocity is zero and such a point is called

stagnation point (see point A above).

STREAMTUBES:



Streamtube

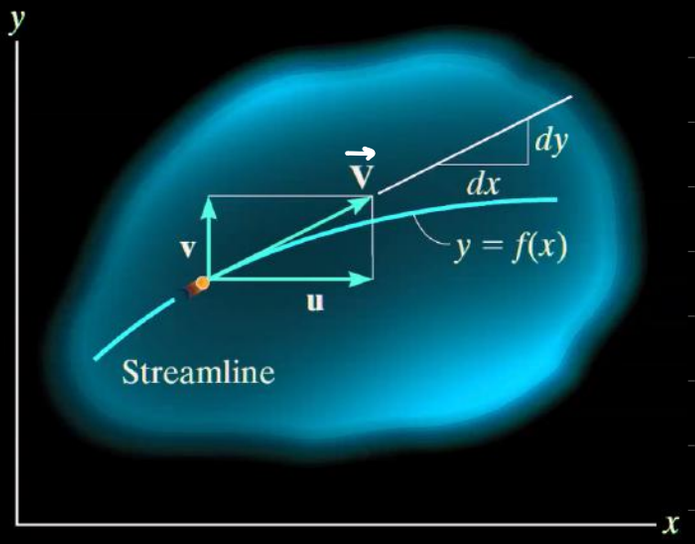


Fluid flows faster within a narrow streamtube

A bundle of streamlines that surrounds a given region of flow forms the streamtube. In case of laminar (streamlined) flow, the fluid flows through streamtube as if it were actually a curved pipe / conduit. The place where streamtube is narrow, the fluid flows faster. (This follows directly from volume conservation).

EQUATION OF A STREAMLINE:

Consider a portion of a stream line in the x-y plane. The local velocity vector is always directed along stream line. Thus by resolving velocity vector in to x and y components, it can be readily seen that $\frac{dy}{dx} = \frac{v}{u}$ (1) where u and v are x and y components of velocity respectively. If we know the velocity field, we can easily find the equation of any streamline by integrating equation (1) and setting the arbitrary constant to satisfy the point through which streamline is required.



The velocity is always tangent to the streamline

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EXAMPLE: A steady two dimensional velocity field is given as $\vec{V} = (0.5 + 0.8x)\hat{i} + (1.5 - 0.8y)\hat{j}$. Find (a) stagnation point (b) Family of streamlines.

Solution: At stagnation point, velocity is zero thus stagnation point can be found by setting x and y components of velocity to zero. This gives us stagnation point as $(-0.625, 1.875)$ m.

(b) Differential equation of streamlines is given as

$$\frac{dy}{dx} = \frac{v}{u} = \frac{1.5 - 0.8y}{0.5 + 0.8x}$$

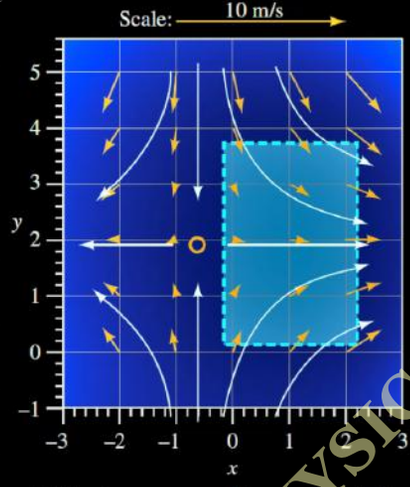
$$\Rightarrow \frac{dx}{0.5 + 0.8x} = \frac{dy}{1.5 - 0.8y}$$

$$\Rightarrow \frac{1}{0.8} \ln(0.8x + 0.5) + \frac{1}{0.8} \ln(-0.8y + 1.5) = C$$

$$\text{or } (0.8x + 0.5)(-0.8y + 1.5) = C$$

i.e

$$\left(x + \frac{5}{8}\right)\left(y - \frac{15}{8}\right) = C$$



Velocity vectors (blue arrows) for the velocity field of Example. The scale is shown by the top arrow, and the solid black curves represent the approximate shapes of some streamlines, based on the calculated velocity vectors. The stagnation point is indicated by the blue circle. The shaded region represents a portion of the flow field that can approximate flow into an inlet.

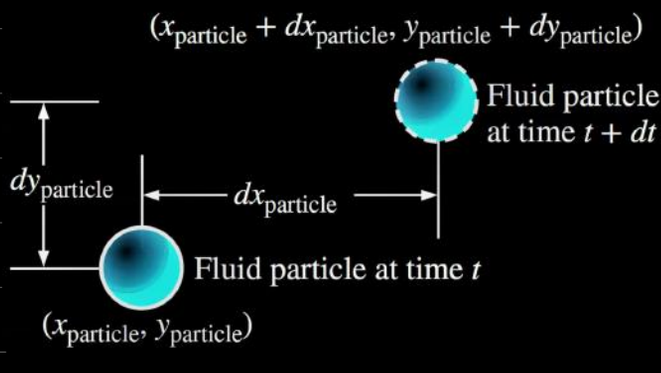


Flow field near the bell mouth inlet of a hydroelectric dam; a portion of the velocity field of Example may be used as a first-order approximation of this physical flow field. The shaded region corresponds to that of **Streamplot**.

This represents family of rectangular hyperbolae and it is plotted in figures above. (This represents a valid family of streamlines to an inlet of a hydroelectric dam. It can be shown that this family satisfies mass conservation law).

PARTICLE ACCELERATION IN FLUID FLOW: (MATERIAL ACCELERATION)

Let $\vec{V}(x, y, t)$ represent velocity of a fluid flow field (2 dimensional). Now suppose a particle goes from (x, y) to $(x + dx, y + dy)$ in a time dt then its change in velocity vector is given as:



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$d\vec{V} = \frac{\partial \vec{V}}{\partial t} dt + \frac{\partial \vec{V}}{\partial x} dx + \frac{\partial \vec{V}}{\partial y} dy$ thus its acceleration is given as: $\vec{a} = \frac{d\vec{V}}{dt} = \frac{\partial \vec{V}}{\partial t} + \frac{\partial \vec{V}}{\partial x}\frac{dx}{dt} + \frac{\partial \vec{V}}{\partial y}\frac{dy}{dt}$

Similarly in case of a 3D flow field,

$$\vec{a} = \frac{\partial \vec{V}}{\partial t} + \frac{\partial \vec{V}}{\partial x}\frac{dx}{dt} + \frac{\partial \vec{V}}{\partial y}\frac{dy}{dt} + \frac{\partial \vec{V}}{\partial z}\frac{dz}{dt} \quad \text{i.e.}$$

$$\vec{a} = \frac{\partial \vec{V}}{\partial t} + u\frac{\partial \vec{V}}{\partial x} + v\frac{\partial \vec{V}}{\partial y} + w\frac{\partial \vec{V}}{\partial z} \quad (1)$$ where u, v and w are x, y and z components of velocities respectively. Now recall that we had defined the del operator (nabla) as $\vec{\nabla} = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$

also $\vec{V} = u\hat{i} + v\hat{j} + w\hat{k}$ thus we can compactly write equation (1) as:

$$\boxed{\vec{a} = \frac{\partial \vec{V}}{\partial t} + (\vec{V}\cdot\vec{\nabla})\vec{V}}$$ ← Material acceleration.

→ In case of steady flow $\partial \vec{V}/\partial t = 0$ however still particles are accelerated due to contribution from second term. Don't get confused here. The velocity at a given point in tube is not changing with time, but particles may go faster or slower as they reach some other point in tube.

EXAMPLE: A steady two dimensional velocity flow field is given as $\vec{V} = (0.5 + 0.8x)\hat{i} + (1.5 - 0.8y)\hat{j}$ calculate material acceleration field.

Solution: $\vec{a} = \frac{\partial \vec{V}}{\partial t} + u\frac{\partial \vec{V}}{\partial x} + v\frac{\partial \vec{V}}{\partial y} = 0 + (0.5 + 0.8x)0.8\hat{i} + (1.5 - 0.8y)(-0.8)\hat{j}$

$\Rightarrow \vec{a} = (0.64x + 0.4)\hat{i} + (0.64y - 1.2)\hat{j}\ m/s^2$.

If we want we can also plot the acceleration field lines (derive family of acceleration field lines) just as we found velocity field lines.

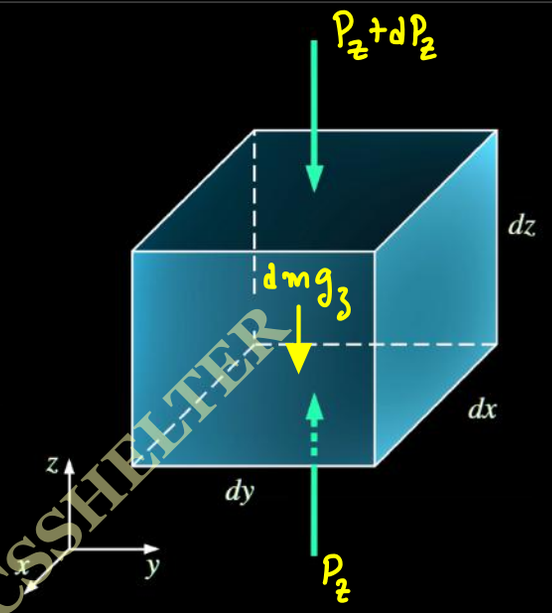
Here the acceleration field family is given by the DE $\frac{dy}{dx} = \frac{a_y}{a_x}$.

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→ Acceleration field can be used to find the resultant of pressure and gravity forces on any fluid particle or volume.

EULER'S EQUATION FOR IDEAL FLUID (NON VISCOUS):

Consider an element of non viscous fluid.
Let us analyse its dynamics in z direction.
Let $\vec{g} = g_x\hat{i} + g_y\hat{j} + g_z\hat{k}$. Now considering the forces in z direction, we can write:



$$-(P_z + dP_z)[dydx] + P_z(dy\,dx) + \rho dxdydz\,g_z$$

$$= \rho\,dxdydz\,a_z$$

Now cancelling terms and dividing by dxdydz, we get:

$$-\frac{dP_z}{dz} + \rho g_z = \rho a_z \quad \text{(Actually here we mean partial derivative)}$$

thus: $-\frac{\partial P_z}{\partial z} + \rho g_z = \rho a_z$ Similarly we can write in x & y directions

thus:

$$-\left[\frac{\partial P_x}{\partial x}\hat{i} + \frac{\partial P_y}{\partial y}\hat{j} + \frac{\partial P_z}{\partial z}\hat{k}\right] + \rho(g_x\hat{i} + g_y\hat{j} + g_z\hat{k}) = \rho(a_x\hat{i} + a_y\hat{j} + a_z\hat{k})$$

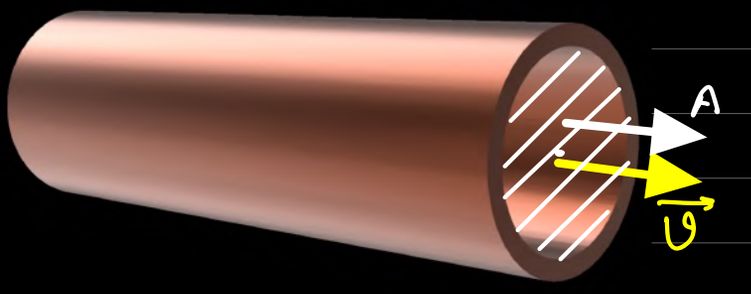
or:

$$\boxed{-\vec{\nabla}P + \rho\vec{g} = \rho\vec{a}}$$

→ This equation can be used for finding pressure field from acceleration field.
→ Here we have not considered any shearing forces because non-viscous liquids are not capable of providing any shearing forces.

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Area Vector And Volumetric Flow Rate:



Consider a pipe whose cross sectional area is A and velocity over its cross section is v then volumetric flow rate is simply $Q = Av$. (see first figure). Now if we were to cut the opening at an angle θ while maintaining same velocity profile, we will still have same volumetric flow rate, however the area of pipe's mouth will now be $A' = A \sec\theta$ (Recall shadow area concept).



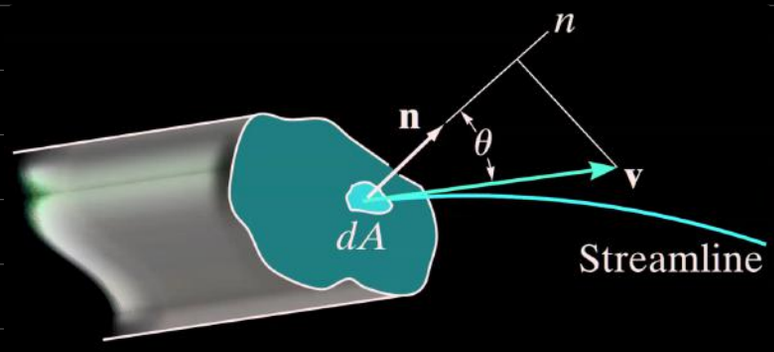
$A' = A \sec\theta$

Thus we can write volumetric flow rate now as $Q = A'\cos\theta\, v$.

Now if we were to define area as a vector with its direction normal to the area patch making an acute angle with velocity we will be able to write flow rate as:

$$Q = \vec{A}' \cdot \vec{v}$$

Now if we have a non uniform velocity field, we can use the same idea to find volumetric flow rate across every infinitesimal stream tube and integrate it to get the net volumetric flow rate. Thus:



$$Q = \oint \vec{v} \cdot d\vec{A}$$

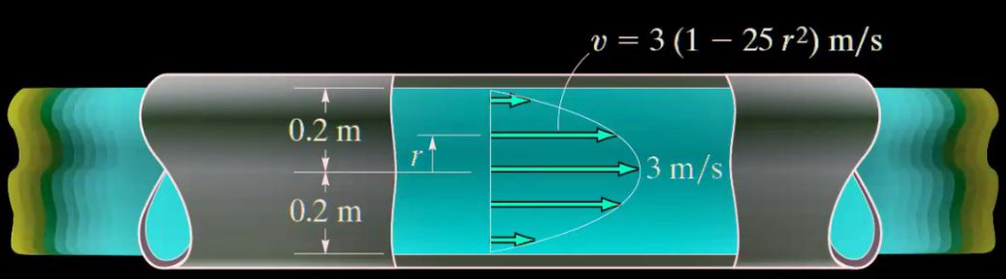
→ In case of non uniform flow, average velocity over a cross section is defined as $v_{avg} = Q/A$.

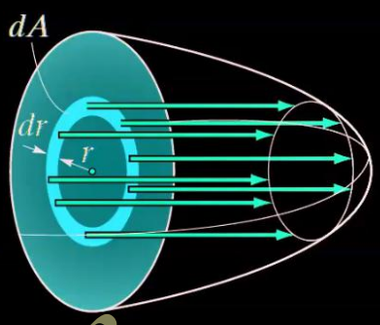
→ This average can't be used for calculating momentum flow rate or kinetic energy flow rate. For that we will have to integrate elemental momentum and energy flow rates.

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Example :

The velocity profile for the steady laminar flow of water through a 0.4-m-diameter pipe is defined by $v = 3(1 - 25r^2)$ m/s, where r is in meters, (Figure). Determine the volumetric flow through the pipe and the average velocity of the flow.





Solution: Here we consider volumetric flow rate through an annulus of radius r and thickness dr. Through this, volumetric flow rate is given as: $dQ = v(r) \times 2\pi r\, dr$ thus:

$$Q = \int_A \mathbf{v} \cdot d\mathbf{A} = \int_0^{0.2\text{ m}} 3(1 - 25r^2)\, 2\pi r\, dr$$

$$= 6\pi \left[\frac{r^2}{2} - \frac{25r^4}{4} \right]_0^{0.2\text{ m}}$$

$$= 0.1885\text{ m}^3/\text{s} = 0.188\text{ m}^3/\text{s} \qquad \textit{Ans.}$$

To avoid this integration, Q can also be determined by calculating the *volume* under the "velocity paraboloid" of radius 0.2 m and height 3 m/s. It is

$$Q = \frac{1}{2}\pi r^2 h = \frac{1}{2}\pi(0.2\text{ m})^2\,(3\text{ m/s}) = 0.188\text{ m}^3/\text{s} \qquad \textit{Ans.}$$

The average velocity is determined as:

$$V = \frac{Q}{A} = \frac{0.1885\text{ m}^3/\text{s}}{\pi(0.2\text{ m})^2} = 1.5\text{ m/s} \qquad \textit{Ans.}$$

→ Recall that volume of a paraboloid is half the volume of its enveloping cylinder i.e $\frac{1}{2}\pi r^2 h$.

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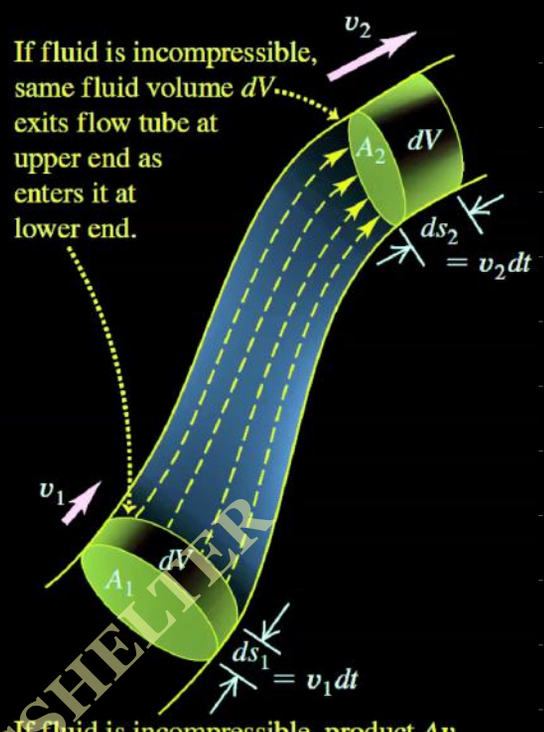
CONTINUITY EQUATION (INCOMPRESSIBLE FLUID):

If there is no accumulation of an incompressible fluid (such as in a pipe with rigid walls), the volumetric inflow rate must be equal to the volumetric outflow rate. Now consider the adjacent figure. Here v_1 is the velocity at the inlet of a stream tube where cross section area is A_1, similarly at outlet the flow velocity and cross sectional areas are v_2 and A_2 respectively. The volume entering the tube in a time dt, $dV = A_1 v_1 dt$ similarly exiting volume in this time, $dV = A_2 v_2 dt$

Thus by conservation of mass:

$$A_1 v_1 dt = A_2 v_2 dt \quad \text{or}$$

$$Q = \boxed{A_1 v_1 = A_2 v_2} \leftarrow \text{continuity equation.}$$



→ If the velocity over a cross section is not uniform, we can apply this equation on every infinitesimal stream tube.

IDEAL FLOW: In general fluid flow can become very complicated however in JEE syllabus we will be focusing on some very simple flows known as ideal flows. The ideal flows have following characteristics:

→ The fluid is non-viscous

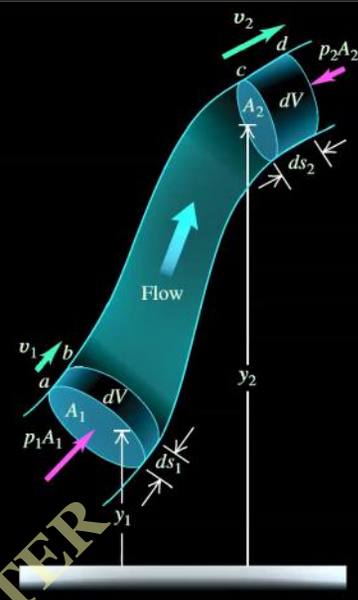
→ The flow is laminar

→ The fluid is incompressible

→ The flow is irrotational (i.e. $\oint \vec{v} \cdot d\vec{l} = 0$ along any contour or $\vec{\nabla} \times \vec{v} = 0$). This means that if a paddle wheel is placed anywhere in the flow, it will not rotate.

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BERNOULLI'S EQUATION: (IDEAL FLOW):



Bernoulli's equation relates the pressure, velocity and height at various points in a streamtube. It is based on work energy theorem (or conservation of Energy if work is also considered as a form of energy). Consider steady flow of an incompressible liquid in a duct. We consider the volume of liquid as it moves infinitesimally in the duct under action of various forces. Now we will find the work done by pressure and gravity forces on this fluid element and equate it to change in its KE (see figure). Now in a time dt, work done by pressure forces:

$$dW_p = P_1 A_1 v_1 dt - P_2 A_2 v_2 dt = (P_1 - P_2) Q dt \quad (1) \text{ where } Q = A_1 v_1 = A_2 v_2.$$

Now we can find work done by gravity by considering the change in gravitational PE, which is as if a mass $\rho A_1 v_1 dt$ disappeared from inlet and appeared at the outlet. Thus work done by gravity:

$$dW_g = -dU_g = -[\rho Q dt\, g\, [y_2 - y_1]) = -\rho Q dt\, g\, [y_2 - y_1]. \quad (2)$$

Now change in Kinetic Energy can also be found by considering the volume "disappeared" from inlet and appeared at the outlet. Thus

$$dKE = \frac{1}{2} \rho Q dt\, [v_2^2 - v_1^2] \quad (3)$$ Now by work energy theorem:

$$dW_p + dW_g = dKE \text{ thus}$$

$$(P_1 - P_2) Q dt + \rho Q g dt (y_1 - y_2) = \frac{1}{2} \rho Q dt\, [v_2^2 - v_1^2]$$

Now cancelling $Q dt$ throughout:

$$(P_1 - P_2) + \rho g (y_1 - y_2) = \frac{1}{2} \rho (v_2^2 - v_1^2) \quad (4)$$

Notice that this equation is nothing but work energy theorem per unit volume. If there were any other field forces doing work, we would have considered their work also in LHS (For instance pseudo force in translationally accelerated frame or centrifugal force in rotating frames).

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Rearranging ④ we get:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 \quad \text{on}$$

between any two points lying on a streamline:

$$\boxed{P + \frac{1}{2}\rho v^2 + \rho g y = \text{Constant}}$$ ← Bernoulli's theorem for incompressible ideal flow.

Alternate Derivation Of Bernoulli's Theorem:

Consider a small fluid element of cross sectional area dA and length ds.

Applying Newton's law along streamline:

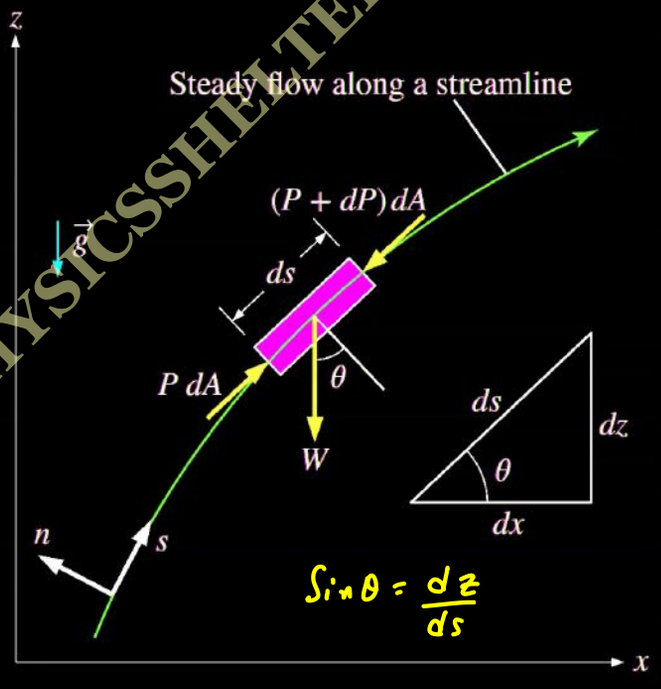
$$P\,dA - (P + dP)\,dA - W\sin\theta = m v \frac{dv}{ds}$$

$$\Rightarrow -dP\,dA - \rho g\,ds\,dA\,\frac{dz}{ds} = \rho\,dA\,ds\,v\frac{dv}{ds}$$

Cancelling dA from each term and simplifying:

$$-dP - \rho g\,dz = \rho v\,dv$$

$$\Rightarrow \boxed{\frac{dP}{\rho} + \frac{1}{2}d(v^2) + g\,dz = 0} \quad ①$$



→ Note that since flow is steady we did not consider $\partial v/\partial t$ terms while writing material acceleration.

→ While writing equation ① we nowhere used incompressibility of the fluid. Thus equation ① is valid for steady compressible as well as incompressible flows.

Thus for arbitrary steady flows along any streamline we can write

$$\boxed{\int \frac{dP}{\rho} + \frac{v^2}{2} + g z = \text{Constant}} \quad ②$$ ← Bernoulli for arbitrary steady flows.

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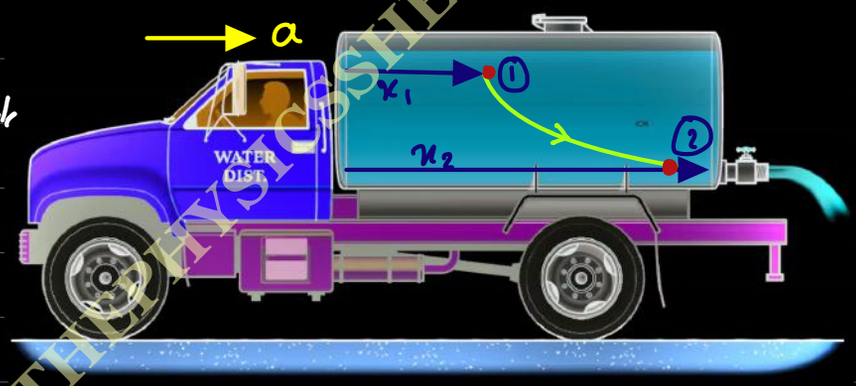
STEADY INCOMPRESSIBLE FLOWS: In case of incompressible flows, ρ is constant and accordingly equation ② gets modified to:

$$\frac{P}{\rho} + \frac{v^2}{2} + gz = \text{Constant} \quad \text{or,}$$

$$P + \frac{1}{2}\rho v^2 + \rho g z = \text{Constant.}$$

Notice that this is exactly the same form of Bernoulli as we derived earlier using work energy theorem.

BERNOULLI'S EQUATION IN HORIZONTALLY ACCELERATED VESSELS:



Let us revisit our usual Bernoulli's equation for incompressible steady flow:

$(P_1 - P_2) = \rho g (y_2 - y_1) + \frac{1}{2}\rho(v_2^2 - v_1^2)$

This equation is nothing but work energy theorem per unit volume.

$P_1 - P_2$ is work done by pressure forces per unit volume,

$\rho g (y_2 - y_1)$ is change in potential energy per unit volume and

$\frac{1}{2}\rho(v_2^2 - v_1^2)$ is change in kinetic energy per unit volume.

Now if the duct carrying the fluid is accelerated, the fluid will also experience a pseudo force and accordingly we will also have to account for work done per unit volume by pseudo force.

Now imagine a vessel accelerated along x axis with an acceleration a. Now if a fluid element of mass dm moves along a streamline (see figure) the work done by pseudo force in moving from point ① to ② along the streamline is $-dma(x_2 - x_1)$ while the volume of the element is $dV = dm/\rho$. Accordingly work done by pseudo force per unit volume is $dW_p = -\rho a(x_2 - x_1)$. Thus Bernoulli's equation after accounting for pseudo force becomes:

$$(P_1 - P_2) - \rho a(x_2 - x_1) = \frac{1}{2}\rho(v_2^2 - v_1^2) + \rho g(h_2 - h_1)$$

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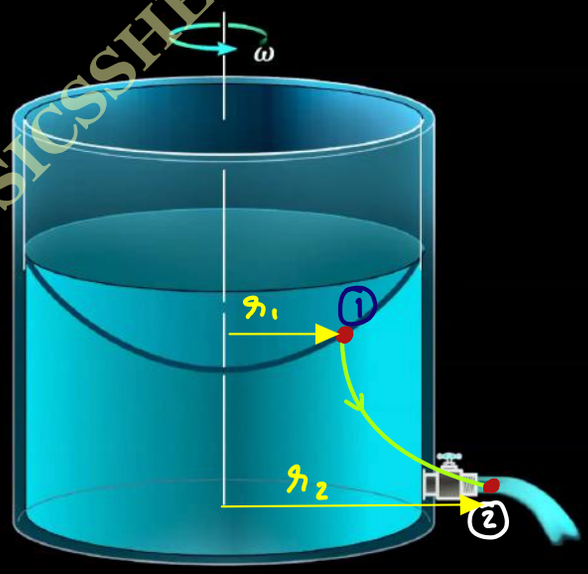
On rearranging we get:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 + \rho a x_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 + \rho a x_2$$

or $$\boxed{P + \frac{1}{2}\rho v^2 + \rho g h + \rho a x = \text{Constant}} \quad (4)$$

→ In case the vessel has a vertical component of acceleration also (say a_y) then g must be replaced by $g + a_y$ in equation (4).

Bernoulli's Equation In Rotating Vessels:



Just as we considered work done by pseudo force per unit volume in case of translationally accelerated vessels, for rotating vessels, we need to consider the work done by centrifugal forces per unit volume. Alternatively we could also talk in terms of centrifugal potential energy per unit volume which is negative of work done by centrifugal forces per unit volume.

Our usual bernoulli's equation is:

$(P_1 - P_2) = \frac{1}{2}\rho(v_2^2 - v_1^2) + \rho g(y_2 - y_1)$. Recall that centrifugal potential energy of any point mass is $-\frac{1}{2} m\omega^2 r^2$. Now consider a mass dm moving from point (1) to point (2) along a streamline (see figure). Change in its centrifugal PE in the process is then,

$dU_c = -\frac{1}{2} dm\,\omega^2(r_2^2 - r_1^2)$. Its volume is dm/ρ. Thus change in centrifugal PE per unit volume is then $-\frac{1}{2}\rho\omega^2(r_2^2 - r_1^2)$. Accordingly our Bernoulli's theorem gets modified as:

$$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) + \rho g(y_2 - y_1) - \frac{1}{2}\rho\omega^2(r_2^2 - r_1^2)$$

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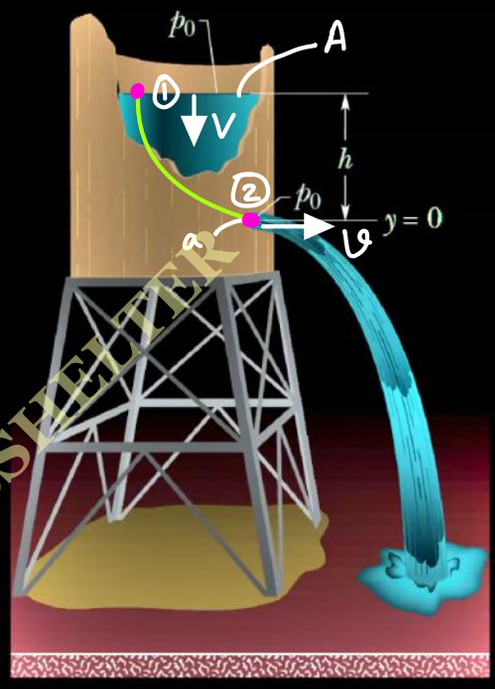
on $$\boxed{P + \frac{1}{2}\rho v^2 + \rho g y - \frac{1}{2}\rho \omega^2 r^2 = \text{Constant}}$$

→ Note that here v is the velocity in the rotating frame.

TORRICELLI'S LAW:

If a small hole is made in an open vessel of wide cross section at a depth h below the free surface of the liquid then the efflux velocity is given as:

$$\boxed{v \approx \sqrt{2gh}}$$



PROOF: Let A be the cross sectional area of the vessel and let a be the area of the hole. Let V be the the velocity of the surface particles and v be the efflux velocity then:

By continuity equation:

$$AV = av \quad (1)$$

Now applying Bernoulli between points 1 and 2:

$$P_0 + \frac{1}{2}\rho V^2 + \rho g h = P_0 + \frac{1}{2}\rho v^2 + 0 \quad (2) \quad (\because \text{Reference is at hole level}).$$

→ Whenever a stream is freely falling (not surrounded by any conduit), pressure inside the stream is atmospheric).

Now solving (1) & (2) we get

$$\boxed{v = \sqrt{\frac{2gh}{1 - (a/A)^2}}}$$

Now if $a << A$, $\quad v \approx \sqrt{2gh}$ QED.

SURFACE ACCELERATION IN PRISMATIC VESSELS:

By continuity equation velocity of surface is given as $V = \frac{a}{A}\sqrt{2gh}$

Now suppose the velocity of surface at initial moment is U

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and after it is displaced through s, its velocity is V then:

$$U = \frac{a}{A}\sqrt{2gh} \quad , \quad V = \frac{a}{A}\sqrt{2g(h-s)}$$

$$\Rightarrow \quad V^2 - U^2 = -\left(\frac{a}{A}\right)^2 \cdot 2gs = 2ws \quad \text{where } w \text{ is acceleration of surface}$$

Thus $$\boxed{w = -\left(\frac{a}{A}\right)^2 g}$$

Thus surface of a prismatic vessel with a hole moves down at a constant deceleration.

EXAMPLE: A completely filled prismatic tank has a small hole at its bottom. It takes 1 hour to completely empty the tank through the hole. How long will it take to drain out 50% of the liquid?

Solution:

$$T = s/[(U+V)/2] \quad \text{thus if } h \text{ is the total height then:}$$

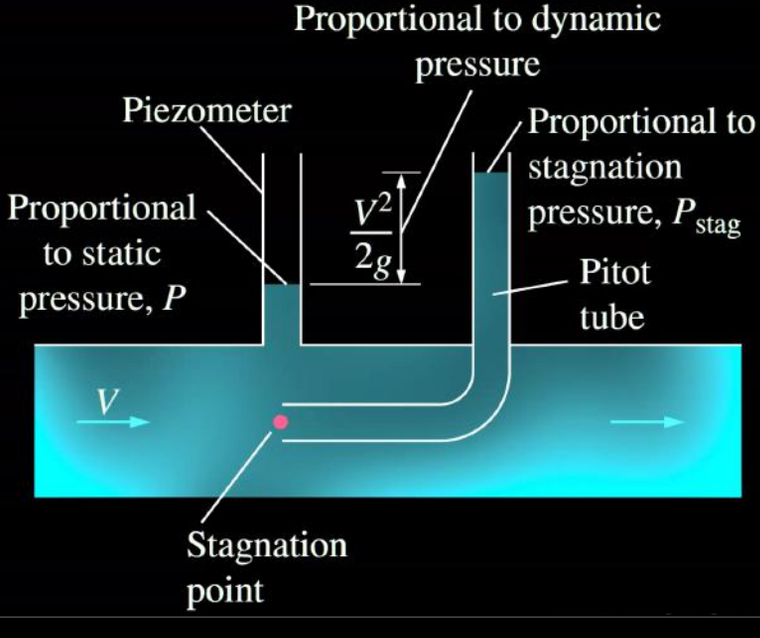
$$1 \text{ hour} = h/\frac{a}{A}\left[\left(\sqrt{2gh}+0\right)/2\right] = \sqrt{2h/g}\ (A/a)$$

$$t = \left[h/2\right]/\left[\left(\frac{a}{A}\right)\left[\sqrt{2gh}+\sqrt{gh}\right]/2\right] = \frac{A}{a}\sqrt{\frac{h}{g}} \cdot \left[\frac{1}{\sqrt{2}+1}\right]$$

$$\Rightarrow \frac{t}{1\text{hour}} = \frac{1}{2+\sqrt{2}} \Rightarrow t = \left(\frac{1}{2+\sqrt{2}}\right)\text{hours} = \frac{2-\sqrt{2}}{2}\text{ hours} \approx 0.29 \text{ hour} = 17.6 \text{ minutes}$$

STATIC, DYNAMIC AND STAGNATION PRESSURES:

PIEZOMETER: A tube exposed to fluid flow with its opening perpendicular to the fluid velocity is called Piezometer (see figure). A piezometer does not obstruct the fluid flow (or cause stagnation) and can be used for measuring pressure of a flowing fluid in any duct.



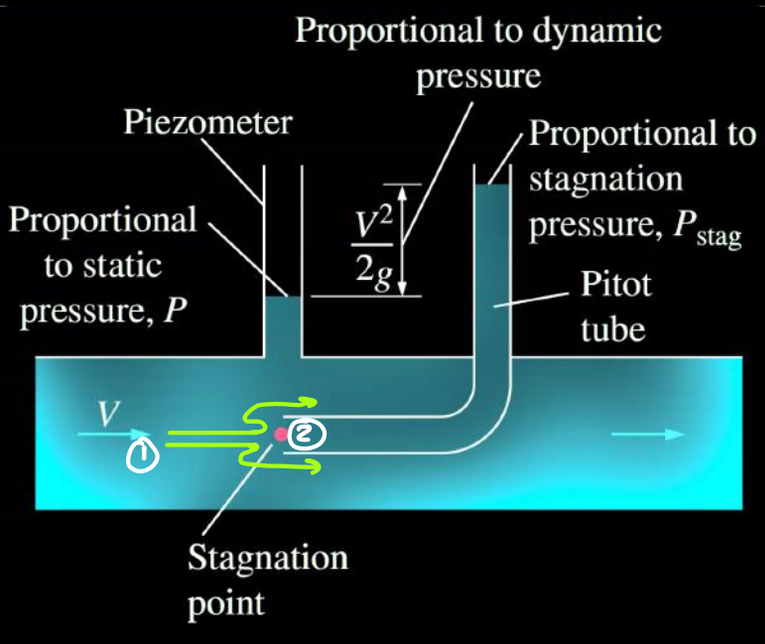
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PITOT TUBE: A bent tube inserted in a fluid flow with its opening perpendicular to the fluid flow is called a Pitot's tube (see figure). A Pitot tube causes stagnation of the fluid and accordingly it can be used to measure pressure of the fluid when it is brought to rest.

STATIC PRESSURE: It represents actual thermodynamic pressure of a fluid as defined in thermodynamics. If h_p is the height of liquid in piezometer then the static gauge pressure is given as:
$P = \rho g h_p$.

DYNAMIC PRESSURE: The term $\frac{1}{2}\rho v^2$ is called dynamic pressure of the fluid. It also represents the change in pressure of the fluid when it is brought to a stop (isentropically).

STAGNATION PRESSURE: The sum of static pressure and dynamic pressure is called the stagnation pressure. This also represents pressure at a point where fluid is brought to rest (isentropically). Mathematically $P_{stag} = P + \frac{1}{2}\rho v^2$. Stagnation pressure can be conveniently measured using a Pitot's tube (see figure). Suppose a Piezometer is fitted on the surface of a flowing fluid and a pitot's tube is inserted just below the surface (such that openings of Pitot's tube and Piezometer are at same level), then pressure far away from Pitot's tube's opening is P. Now applying bernoulli's equation between points ① and ②: $P + \frac{1}{2}\rho v^2 = P_{stag}$



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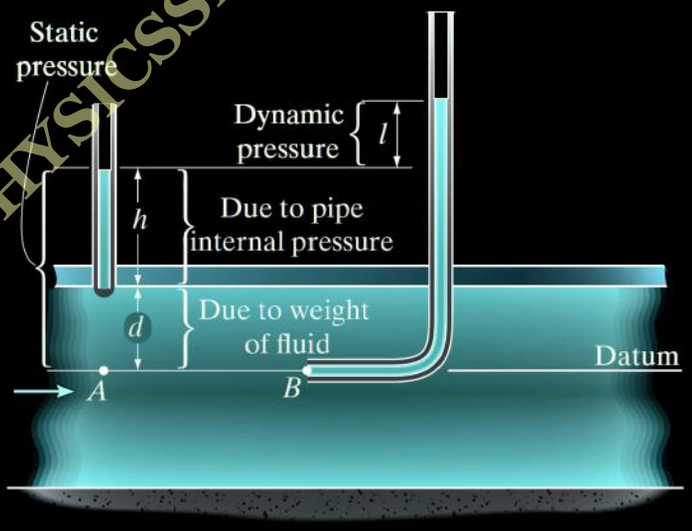
Thus $$V = \sqrt{\frac{2\,(P_{stag} - P)}{\rho}}$$

Now $P = \rho g h_{stat}$ and $P_{stag} = \rho g h_{stag}$ where h_{stat} and h_{stag} are heights in Piezometer and Pitot's tube thus,

$$V = \sqrt{2g\,\Delta h}$$ where Δh is height difference between pitot's tube and piezometer.

HYDROSTATIC PRESSURE:

Pressure due to weight of the liquid is called hydrostatic pressure. In a steady flow if the fluid is not vertically accelerated, then hydrostatic pressure can be added just as it is added in fluids confined in vessels. For example, in adjacent figure if Pressure reading of Piezometer is P then pressure at A is simply $P + \rho g d$ where we can called $\rho g d$ as hydrostatic pressure term. Hydrostatic pressure term depends on reference level.



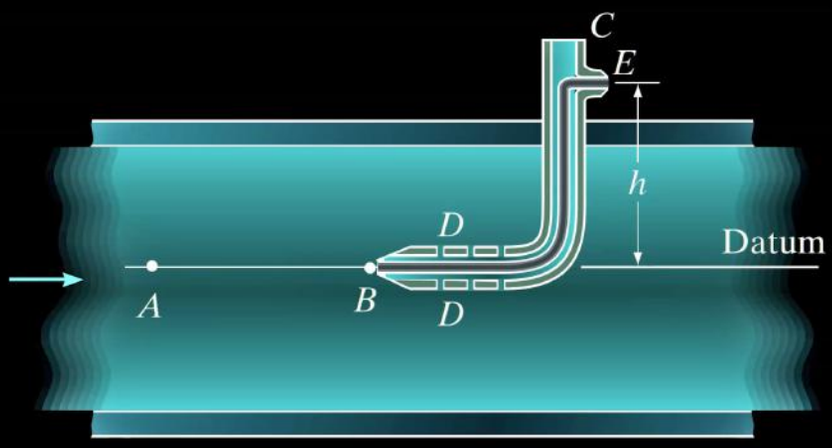
Piezometer and pitot tubes

TOTAL PRESSURE: Sum of static, dynamic and hydrostatic pressures at any point is called the total pressure at any point.

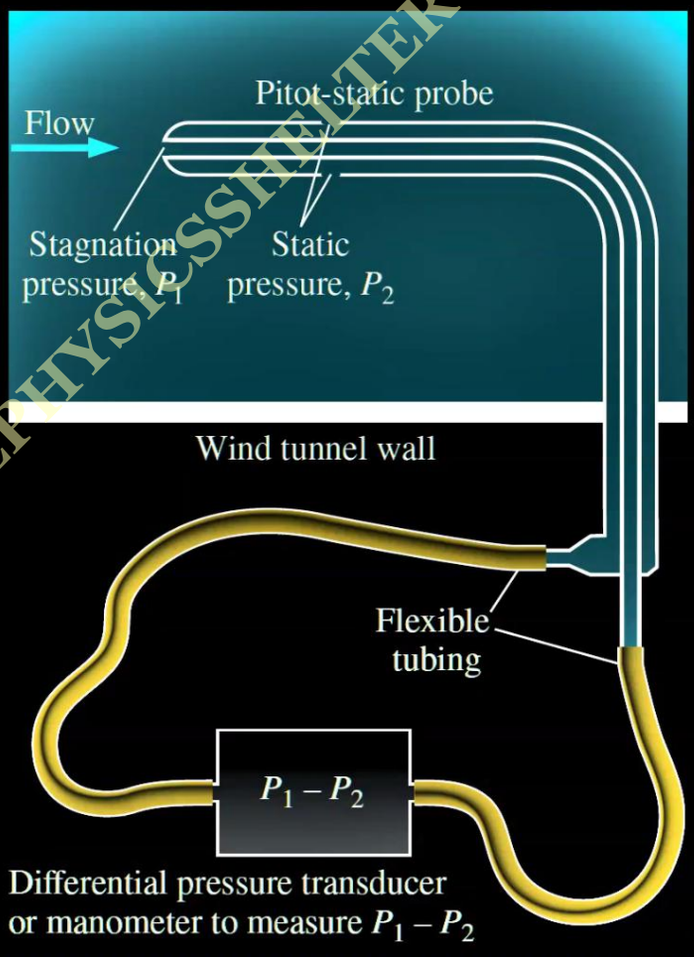
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Pitot Static Tube :

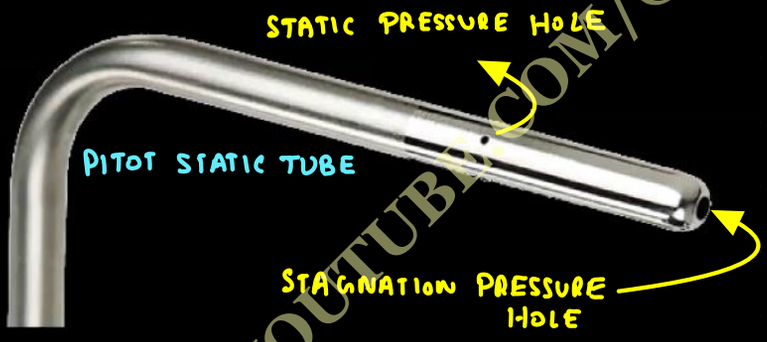
This is a device from which we can readily measure the dynamic pressure (hence flow velocity) of a fluid. It consists of concentric tubes as shown in the figure. One of the tube's openings faces the velocity "head on" while, outer tube's openings are perpendicular to velocity of the flow. Thus a manometer connected to point E will give us the stagnation pressure and that connected to C will give us the static pressure and the difference of heights will give us the dynamic pressure (and hence flow velocity).



Pitot-static tube



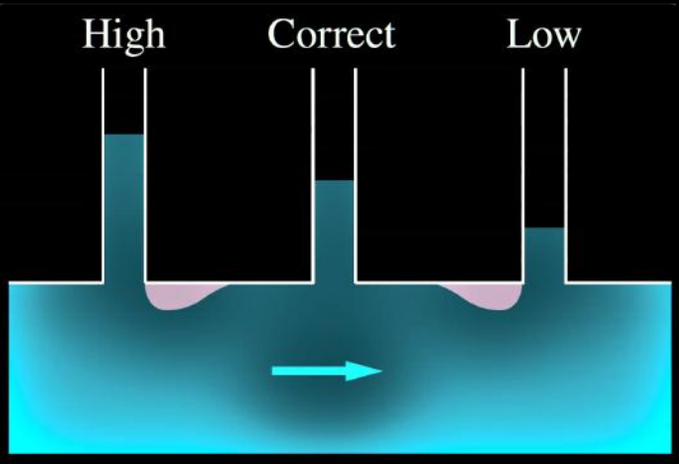
Differential pressure transducer or manometer to measure $P_1 - P_2$



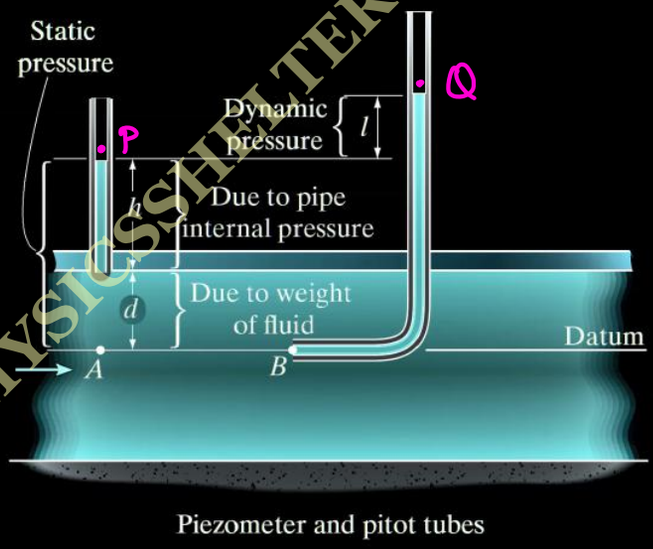
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Alternatively, the outlet of two tubes may be connected to a differential pressure transducer which directly reads the dynamic pressure i.e difference of stagnation and static pressures.

→ Static Pressure taps must be drilled carefully, otherwise we may end up getting wrong reading for static pressure due to obstacles around piezometer mouth (see figure). An obstacle may slow down or speed up the local flow thus causing errors.



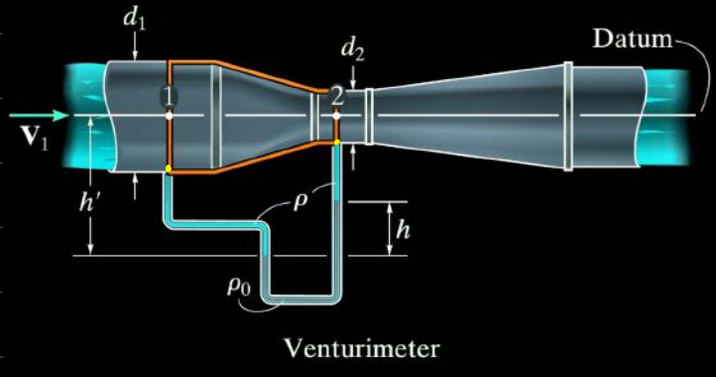
→ In general Bernoulli's theorem can be applied between two points only if they lie on the same streamline, otherwise we may end up getting absurd results. For instance in adjacent figure we can't apply Bernoulli's theorem between P and Q, otherwise we will get: $P_0 + \rho g h = P_0 + \rho g (h+l)$ or $l = 0$ which is obviously absurd. (Btw presence of tubes makes the flow rotational, for example near the mouth of piezometer, we can find a path where $\int \vec{v} \cdot d\vec{l} \neq 0$.) Bernoulli's theorem can be applied across points on different streamlines only if the flow is irrotational).



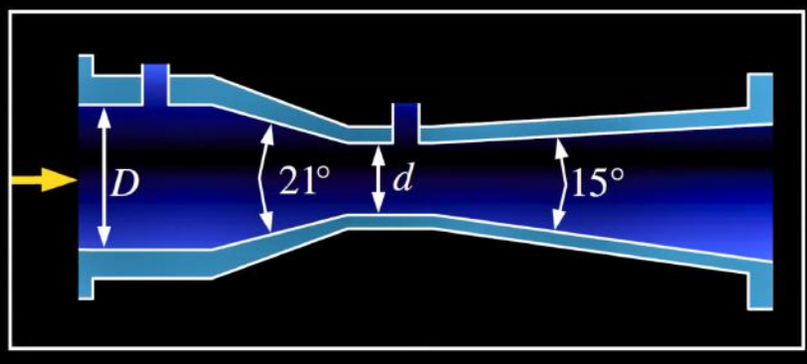
Piezometer and pitot tubes

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VENTURIMETER:



Venturimeter



A venturimeter is a device used for measurement of flow rate through a conduit. It is essentially a pipe like structure with converging and diverging sections with outlets for connecting manometer limbs (see figure). Usually mercury is used as a working fluid in the manometer (say its density is ρ_0). Part of the tubes also gets filled with the material of the flowing fluid (density ρ). We will try to develop an expression for flow velocity in terms of difference in heights of mercury in the two columns. Let V_1 and d_1 be the velocity and diameter at the venturimeter inlet and V_2 and d_2 be the corresponding values at least cross section point. Then by continuity equation:

$$V_1 \times \pi d_1^2/4 = V_2 \times \pi d_2^2/4 \quad (1)$$, Now by Bernoulli's equation:

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2 \quad (2)$$

Solving (1) & (2) we get:

$$V_1 = \sqrt{\frac{2(P_1 - P_2)/\rho}{(d_1/d_2)^4 - 1}}$$

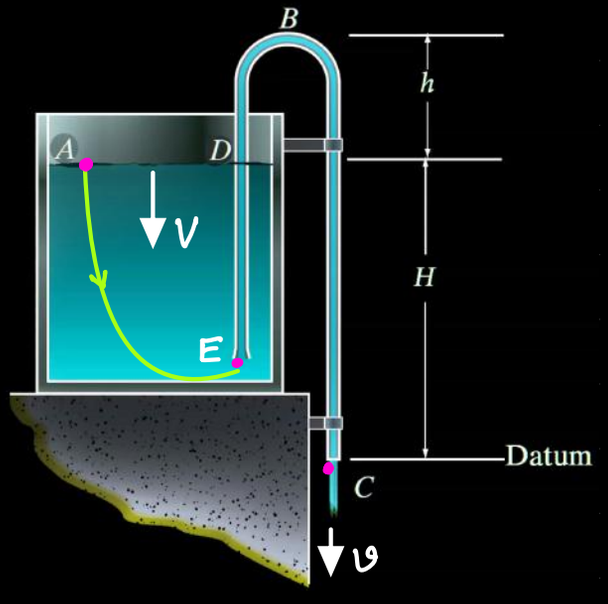
Now from the manometer, $P_1 - P_2 = (\rho_0 - \rho) g h$ thus:

$$\boxed{V_1 = \sqrt{\frac{2(\rho_0/\rho - 1)}{(d_1/d_2)^4 - 1}\, g h}}$$

→ When gases are flowing, typically $\rho_0/\rho \gg 1$ and accordingly we may take $\rho_0/\rho - 1 \approx \rho_0/\rho$.

Siphon:

A siphon is a device used for draining out liquids from tanks. Its outlet is below the upper surface of the tank and liquid drains out until free surface of water remains above the siphon outlet. We will prove it using Bernoulli and Continuity.



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Let cross sectional area of vessel be A and that of siphon outlet be a. Now let us imagine a streamline from surface point A to just outside siphon outlet C via point E. Now by continuity:

$AV = av$ ① (V: Velocity of surface, v: efflux velocity at siphon outlet).

Now using Bernoulli's equation between these two points:

$$P_0 + \frac{1}{2}\rho V^2 + \rho g H = P_0 + \frac{1}{2}\rho v^2 \quad ②$$

Solving ① & ②,

$$v = \sqrt{\frac{2gH}{1-(a/A)^2}} \quad ③$$ ← Siphon Efflux Velocity.

→ Actually both equations are exactly same as the ones we wrote for deriving Torricelli's law, and therefore we get same solution.

→ For a real value of v, $H > 0$ thus a siphon can function only so long as outlet of siphon is below the free surface of water. The depth of siphon tube inside the vessel has no effect on efflux velocity, however if the water surface reaches the lower end of siphon tube inside the vessel, the siphoning will obviously stop.

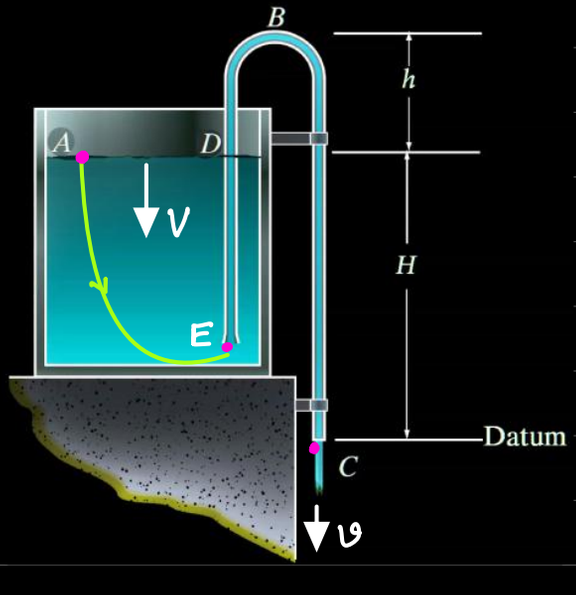
→ Height (h) of the siphon tube has no effect on efflux velocity though for siphon to function $H + h < P_0/\rho g$, if it is more than that, vapour cavity develops in the siphon and it does not function any more. This condition is called Cavitation. (Actually a large value of h causes water to vaporize due to low pressure).

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Minimum Pressure And Cavitation In Siphons:

Using Bernoulli's equation it can be seen that least pressure in a siphon occurs at point B. If this pressure is very low, water will vaporize. Let us assume $a << A$ and apply Bernoulli theorem between B and C:

$$P_0 + \frac{1}{2}\rho V^2 + 0 = P_B + \frac{1}{2}\rho v^2 + \rho g(H+h)$$



$\Rightarrow P_B = P_0 - \rho g(H+h)$, now theoretically the pressure cannot be negative according to our understanding of pressure being due to collisions of molecules. Thus theoretical Height of siphon tube follows the inequality:

$$P_B > 0 \Rightarrow P_0 - \rho g(H+h) > 0$$

$$\Rightarrow \boxed{(H+h) < P_0/\rho g}$$ ← Theoretical max height of limb with downward flow.

Now for water $\rho = 1000\ kg/m^3 \Rightarrow H+h < 1.01\times10^5/(10^3\times9.8) = 10.31\ m$

→ Thus maximum theoretical height of downward limb of siphon can be about 10.31 but practically cavitation will start before the theoretical limit due to reduction in boiling point at low pressures.

→ For the same reason any pump placed on ground can't pull water from a point more than 10.31 metre deep. This is why submersible pumps are commonly used (especially if water table is very low).

Head Form Of Bernoulli's Equation:

Our standard form of Bernoulli's equation is given as:

$P + \frac{1}{2}\rho v^2 + \rho g H = \text{Constant}$. If we wish, we can divide the entire equation by the term ρg (Constant for a given liquid) and we get:

$$P/\rho g + v^2/2g + H = \text{Constant}.$$ This is known as Head form of Bernoulli's equation. The terms are individually known as:

- $P/\rho g$ → Pressure Head
- $v^2/2g$ → Velocity Head
- H → Gravitational Head.
- $P + v^2/2g + H$ → Total Head.

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HEAD LOSS: While deriving Bernoulli's theorem, we assumed no frictional or turbulence losses during the fluid flow but practically there is always some loss of mechanical energy due to these effects. The energy lost per unit weight as the fluid travels from one place to another is called head loss. If h is the head loss then mechanical energy lost due to friction and turbulence is simply mgh, where m is the total mass flown. The fluid may lose energy not only due to frictional effects but also due to mechanical work extracted from the fluid such as in case of turbines (on windmills or watermills etc).

HEAD GAIN: Just as a fluid can lose energy due to frictional losses, it can also gain energy due to work done by a pump. The energy gained by fluid per unit weight due to such a process is called Head gain. Thus if h is the head gain in a pumping process, energy gained by the fluid is mgh where m is the total mass pumped.

MODIFICATION OF BERNOULLI'S EQUATION IN CASE OF HEAD GAIN / LOSS:

Let h be the head loss of a fluid as it travels from point A to point B then using energy conservation principle, we can write:

$$\frac{P_A}{\rho g} + \frac{v_A^2}{2g} + h_A - h = \frac{P_B}{\rho g} + \frac{v_B^2}{2g} + h_B.$$

→ In case of head gain, $-h$ will be replaced by a $+h$ on the LHS of above equation.

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Limitations Of Bernoulli's Theorem:

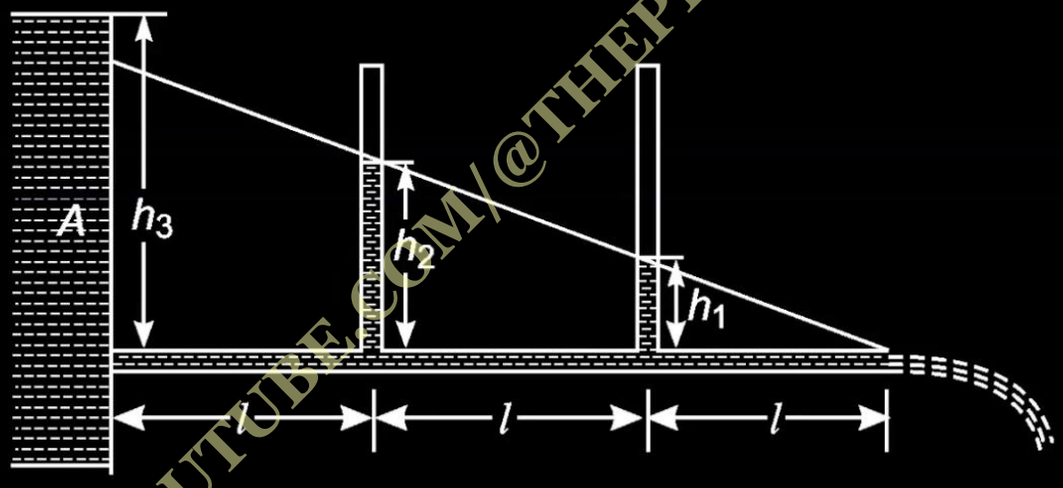
Bernoulli's theorem can be used only when:

1. Flow is steady 2. Flow is non-viscous 3. There is no shaft work (such as in pump or turbine) 4. Flow is incompressible 5. Heat transfer is negligible (Heat changes density) 6. Flow is streamlined (not turbulent).

To overcome these limitations, various modifications are made to standard Bernoulli's theorem such as the one we saw in a previous section to account for head losses and gains.

Example : Irodov: 1.335:

In the arrangement shown in Figure a viscous liquid whose density is $\rho = 1.0\ \text{g/cm}^3$ flows along a tube out of a wide tank A. Find the velocity of the liquid flow, if $h_1 = 10\ \text{cm}$, $h_2 = 20\ \text{cm}$, and $h_3 = 35\ \text{cm}$. All the distances l are equal.



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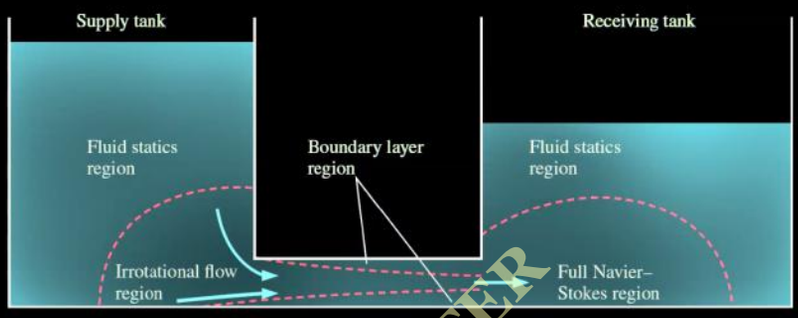
Given: Head loss per unit length in horizontal pipe is constant.
Ignore any Head losses in the tank.

Solution: Head loss in middle one third of horizontal pipe is $h_2 - h_1$, thus net head loss must be $3(h_2 - h_1)$. Now applying modified equation Between vessel top ($V \approx 0$) and exit:

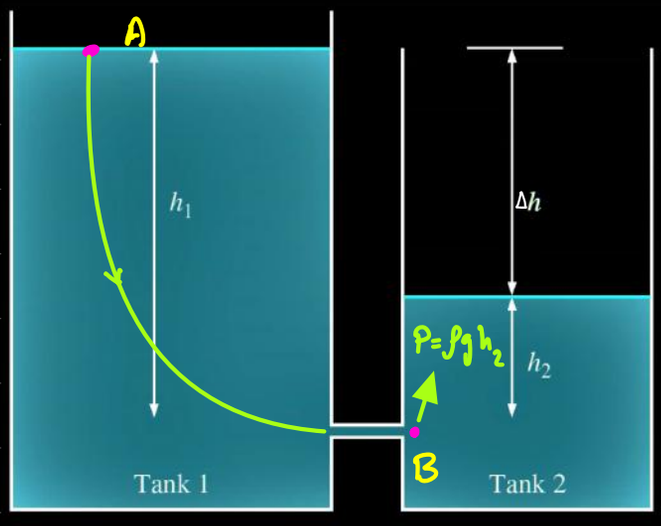
$$\frac{P_0}{\rho g} + \frac{0^2}{2g} + h_3 - 3(h_2 - h_1) = \frac{v^2}{2g} + \frac{P_0}{\rho g} \Rightarrow v = \sqrt{2g\left[h_3 - 3(h_2 - h_1)\right]}$$

$\Rightarrow \quad v = \sqrt{20(35-30)/100} = 1 \text{ ms}^{-1}$ Ans.

Fluid Motion Between Connected Tanks Through A Fine Duct:



For any real fluid, strictly speaking Bernoulli's theorem is never applicable, however when various heads are fairly large as compared to head losses, we get fairly accurate results using Bernoulli's theorem. When two tanks containing a fluid are connected through a conduit, the amount of head loss depends on the diameter of connecting pipe, viscosity, shape of entry mouth (sharp corners cause more losses due to turbulence) and some other geometrical factors. The general analysis of such flow is fairly complicated and involves solution of Navier stoke's partial differential equation which is far beyond the scope of JEE, However when the cross section radius of the connecting tube is very small as compared to the height of liquids in the tanks, we may assume that the pressure at the exit of fine duct is simply the hydrostatic pressure of the recieving vessel. (see point B in adjacent figure). This can also be understood intuitively.



Suppose we discharge some liquid from an injection near the ocean bed, almost instantaneously the pressure at the exit of injection will be same as that before the discharging process started. A small volume being slowly discharged under a very deep ocean does not make any practical difference to the pressure near the ocean bed. Thus for finding instantaneous flow rate in such problems we can apply Bernoulli's equation between points A and B (see figure), while assuming gauge pressure at B to be simply $\rho g h_2$.

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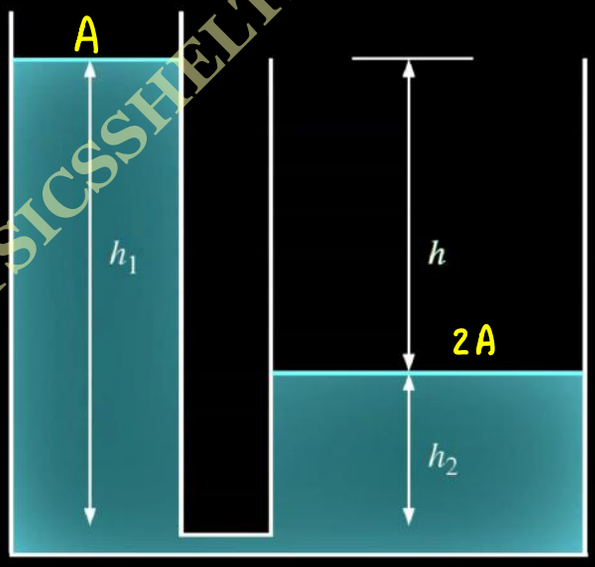
Now since the conduit is very fine, we may assume surface velocity to be nearly zero. Thus Bernoulli equation becomes

$$P_0 + \rho g h_1 + 0 = P_0 + \rho g h_2 + \frac{1}{2}\rho v^2$$

$$\Rightarrow \boxed{v = \sqrt{2g(h_1 - h_2)} = \sqrt{2gh}}$$

→ In this type of flow, there is a head loss due to inelastic collisions of the existing particles with existing particles in the other tank.

EXAMPLE: Two tanks of unequal areas are connected by a conduit of cross sectional area a ($<< A$). Assuming liquid to be non viscous, after how much time will the water level be equalised. The figure shows the configuration of the system at $t = 0$. (Cross sectional areas of vessels are A and $2A$ as shown).



Solution: At shown instant, $v = \sqrt{2gh}$

Thus rate of fall of level in left vessel $= av/A = a\sqrt{2gh}/A$

Rate of rise of level in right vessel $= av/2A = a\sqrt{2gh}/2A$

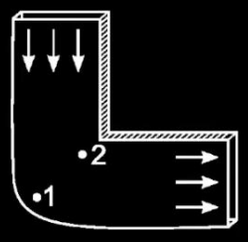
$$\Rightarrow \frac{dh}{dt} = -\frac{a\sqrt{2gh}}{A}\left[\frac{3}{2}\right] \Rightarrow \int_h^0 \frac{dh}{\sqrt{h}} = \frac{-3a}{2A}\sqrt{2g}\int_0^T dt$$

$$\Rightarrow 2\sqrt{h} = \frac{3a}{2A}\sqrt{2g}\,T \Rightarrow T = \frac{2A}{3a}\sqrt{\frac{2h}{g}} \text{ Ans where } h = h_1 - h_2.$$

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EXAMPLE: IRODOV: 1·315:

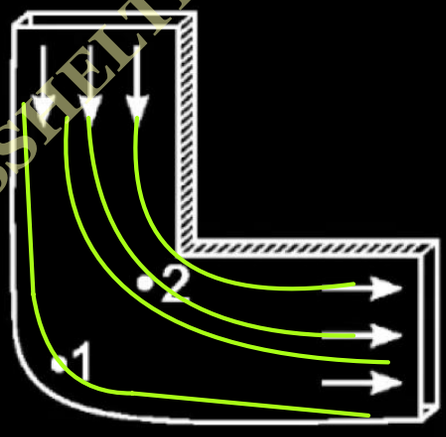
Ideal fluid flows along a flat tube of constant cross-section, located in a horizontal plane and bent as shown in Fig. 1.80 (top view). The flow is steady. Are the pressures and velocities of the fluid equal at points 1 and 2? What is the shape of the streamlines?



Solution: As the particles move across section 1-2, they bend around and hence have an inward centripetal acceleration (i.e from 1 to 2). For this an inward pushing force is required. Thus, $P_1 > P_2$.

Now let for field pressure and velocity be P and v then, for streamline passing through (1):

$$P + \frac{1}{2}\rho v^2 = P_1 + \frac{1}{2}\rho v_1^2 \quad (A)$$



Similarly for streamline passing through 2:

$$P + \frac{1}{2}\rho v^2 = P_2 + \frac{1}{2}\rho v_2^2 \quad (B)$$

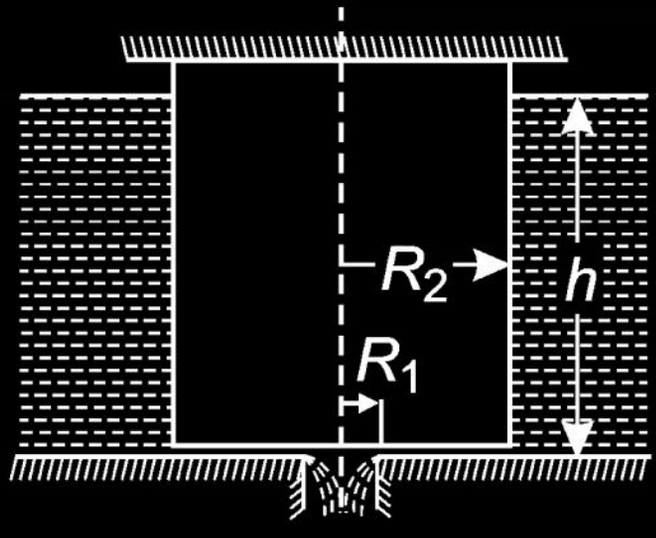
From (A) and (B):

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$ thus,

$\because P_1 > P_2 \therefore v_2 > v_1$ accordingly streamlines near 2 will be closely spaced.

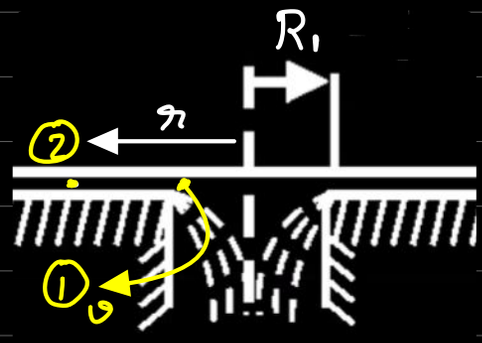
EXAMPLE: IRODOV 1·321:

The horizontal bottom of a wide vessel with an ideal fluid has a round orifice of radius R_1 over which a closed cylinder is mounted whose radius is R_2 ($>R_1$). The clearance between the cylinder and bottom of the vessel is very small, the fluid density is ρ. Find the static pressure of the fluid in the clearance as a function of distance r from the axis of the orifice, if the height of fluid is equal to h.



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Solution: Consider the adjacent figure. At point ①, the pressure is P_0 as water starts its free fall and by Torricelli's law, efflux velocity is $\sqrt{2gh}$ (One can readily see that both Bernoulli and continuity equations between surface and this point are same as the ones used for deriving Torricelli's law).



Note that as water moves in the portion will begin to collide inelastically near the axis and there will be turbulence. Thus in the region of inelastic collisions, Bernoulli equation will not be valid any more. Now let thickness of clearance be b. By continuity between points ① and ②:

$$2\pi r b \times v_2 = 2\pi R_1 b \times \sqrt{2gh} \quad ①$$

, Now using Bernoulli:

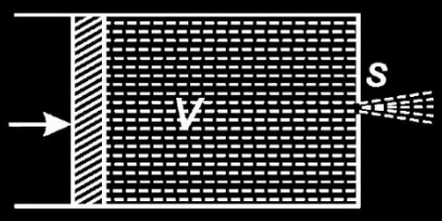
$$P_0 + \frac{1}{2}\rho(\sqrt{2gh})^2 = P + \frac{1}{2}\rho v_2^2 \quad ②$$

Solving ① & ②: $v_2(r) = \sqrt{2gh}\,\frac{R_1}{r}$ Ans.

$$P = P_0 + \rho g h\left[1 - \left(\frac{R_1}{r}\right)^2\right] \text{ Ans.}$$

EXAMPLE : IRODOV 1.322 :

What work should be done in order to squeeze all water from a horizontally located cylinder (See Figure) during the time t by means of a constant force acting on the piston? The volume of water in the cylinder is equal to V, the cross-sectional area of the orifice is s, with s being considerably less than the piston area. The friction and viscosity are negligible small.



Solution: If v is the efflux velocity, then $svt = V$ ①

Now kinetic energy of exiting fluid : $KE = \frac{1}{2}\rho V v^2$ ② this must also be the work done on the fluid thus from ① & ②, required work:

$$DW = \frac{1}{2}\rho V\left(\frac{V}{st}\right)^2 = \frac{1}{2}\frac{\rho V^3}{s^2 t^2} \text{ Ans.}$$

Alternatively : Let F be the pushing force on piston of cross section A

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$(A >> s)$ then using bernoulli:

$$\frac{F}{A} + 0 + 0 = \frac{1}{2}\rho v^2 \quad ①$$

Now if l is the piston displacement, $dW = Fl = \frac{1}{2}\rho v^2 A l$

i.e $dW = \frac{1}{2}\rho V\left(\frac{V}{st}\right)^2 = \frac{1}{2}\frac{\rho V^3}{s^2 t^2}$ Ans.

EXAMPLE: IRODOV 1·324:

A horizontally oriented tube AB of length l rotates with a constant angular velocity ω about a stationary vertical axis OO' passing through the end A (Fig. 1.86). The tube is filled with an ideal fluid. The end A of the tube is open, the closed end B has a very small orifice. Find the velocity of the fluid relative to the tube as a function of the column "height" h.

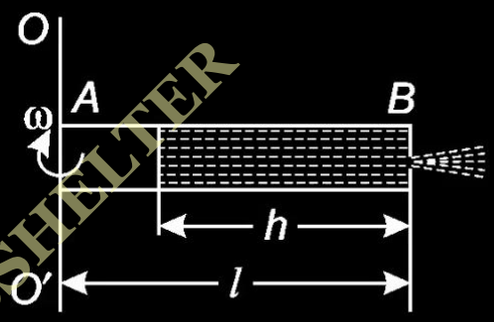
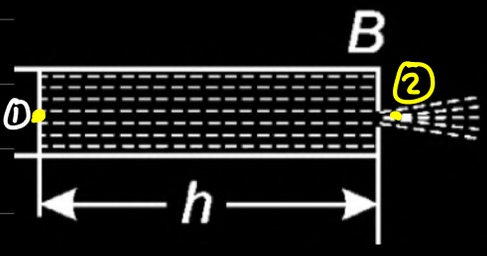


Fig. 1.86

Solution: Here we can directly use Bernoulli's equation for rotating frames i.e $P + \frac{1}{2}\rho v^2 + \rho g z - \frac{1}{2}\rho \omega^2 r^2 = \text{constant}$.

Thus between ① & ② we get:

$$P_0 + \frac{1}{2}\rho \times 0^2 + 0 - \frac{1}{2}\rho\omega^2(l-h)^2 = P_0 + \frac{1}{2}\rho v^2 - \frac{1}{2}\rho\omega^2 l^2 \quad ⓘ$$



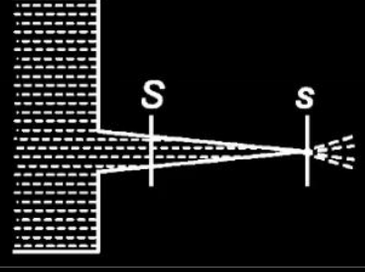
(Since hole is very small we assumed velocity at ① nearly zero). Now solving equation ⓘ:

$$v = \omega\sqrt{h(2l-h)} \quad \text{Ans.}$$

EXAMPLE: IRODOV: 1·329:

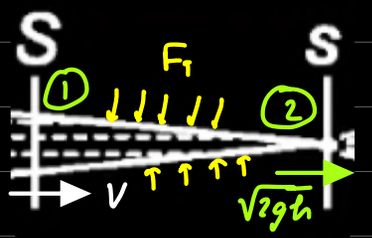
A side wall of a wide open tank is provided with a narrowing tube (Fig. 1.88) through which water flows out. The cross-sectional area of the tube decreases from $S = 3.0\,\text{cm}^2$ to $s = 1.0\,\text{cm}^2$. The water level in the tank is $h = 4.6$ m higher than that in the tube.

Neglecting the viscosity of the water, find the horizontal component of the force tending to pull the tube out of the tank.



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Solution: Consider the free body diagram of portion of water in the tapering tube. Here the rate of change of momentum of water must be due to resultant of gauge pressure force offered by the water behind S and the gauge pressure force exerted by the tapering tube. Now applying Bernoulli between ① & ②



$$P_1 + \frac{1}{2}\rho V^2 = 0 + \frac{1}{2}\rho \times 2gh = \rho g h \Rightarrow P_1 + \frac{1}{2}\rho V^2 = \rho g h \quad (I)$$

(P_1 : Gauge pressure).

By Continuity, $SV = s\sqrt{2gh}$ (II) from (I) & (II):

$$P_1 = \rho g h\left[1 - \frac{s^2}{S^2}\right], \quad V = \sqrt{2gh}\,\frac{s}{S}$$

Now $P_1 S - F_T = \rho s \sqrt{2gh}\left[\sqrt{2gh} - \sqrt{2gh}\,\frac{s}{S}\right]$ (F_T : Tube force).

($\dot{m}(V_{exit} - V_{entry})$).

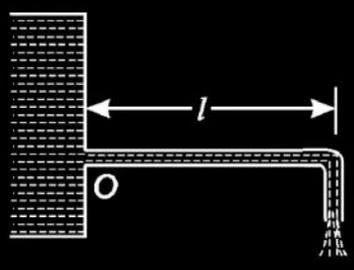
$$\Rightarrow F_T = \rho g h S\left[1 - \frac{s^2}{S^2}\right] - 2\rho s g h\left[1 - \frac{s}{S}\right]$$

$$\Rightarrow F_T = \frac{\rho g h}{S}(S-s)^2 \quad \text{Ans.}$$

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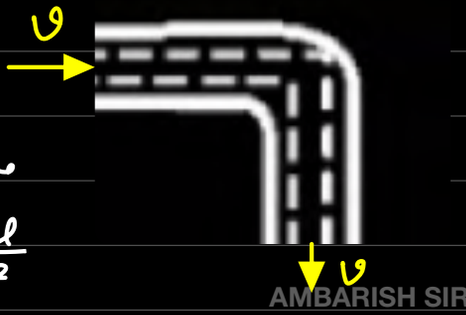
EXAMPLE: IRODOV: 1.328:

Water flows out of a big tank along a tube bent at right angles: the inside radius of the tube is equal to $r = 0.50$ cm (Fig. 1.87). The length of the horizontal section of the tube is equal to $l = 22$ cm. The water flow rate is $Q = 0.50$ litres per second. Find the moment of reaction forces of flowing water, acting on the tube's walls, relative to the point O.



Solution: Here Irodov has ignored the torque due to weight of water in the horizontal pipe considered the effect of bringing about change in angular momentum of water at the bend.

The torque on pipe section is equal and opposite to the torque on water (i.e dL/dt of water).



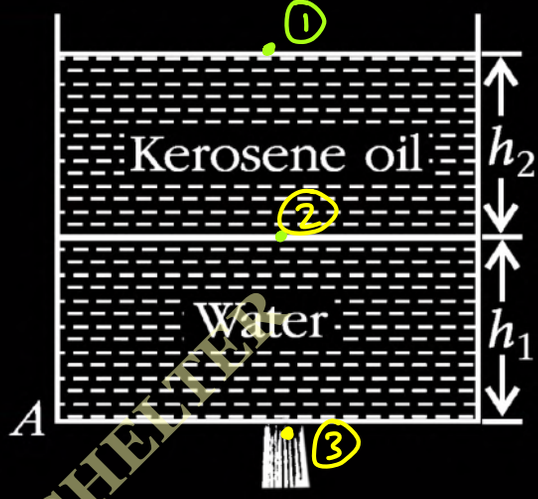
Now rate of creation of angular momentum at the bend, $\frac{dL}{dt} = \rho Q v \times l = \rho Q \cdot \frac{Q}{\pi r^2} l = \frac{\rho Q^2 l}{\pi r^2}$

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This must also be the reaction torque on pipe thus an required moment of reaction forces, $\tau = \rho Q^2 l / \pi r^2$ Ans

EXAMPLE: IRODOV 1.318:

A wide vessel with small hole at the bottom is filled with water and kerosene. Neglecting the viscosity, find the velocity of water flow if the thickness of water layer is $h_1 = 30\,cm$ and that of kerosene layer is $h_2 = 20\,cm$. Take densities of water and kerosene as ρ_w and ρ_k.



Solution: Here since the two liquids are different, we can't directly apply Bernoulli's theorem between points ① and ③. However since vessel has a wide cross section, we may assume nearly zero velocity in the vessel cross section. Accordingly pressure at 2 may be simply considered as the hydrostatic pressure due to kerosene column. Now applying Bernoulli between ② and ③:

$$\rho_k g h_2 + \rho_w g h_1 = \frac{1}{2}\rho_w v^2 \Rightarrow v = \sqrt{\frac{2g(\rho_k h_2 + \rho_w h_1)}{\rho_w}} \quad \text{Ans.}$$

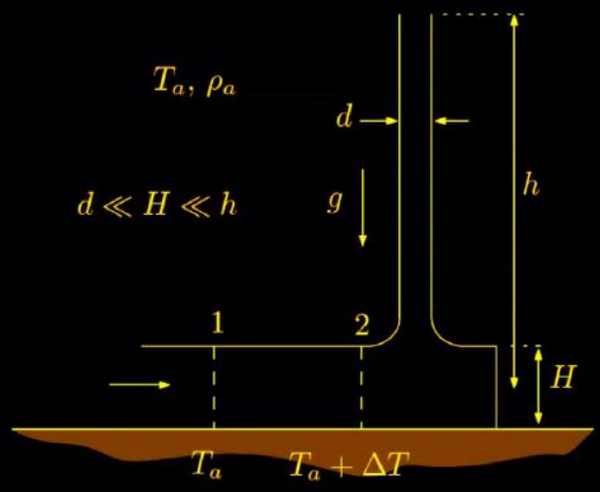
→ Alternatively we could convert kerosene column in to an equivalent column of water for hydrostatic pressure purposes and then use Torricelli's law. Thus $h_{eff} = h_1 + h_2 \rho_k/\rho_w$. Thus using Torricelli's law,

$$v = \sqrt{2g\left[h_1 + \frac{\rho_k h_2}{\rho_w}\right]}$$

This is same result as we got earlier.

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Example: JEE Advanced 2023 : Deleted Question (Corrected):



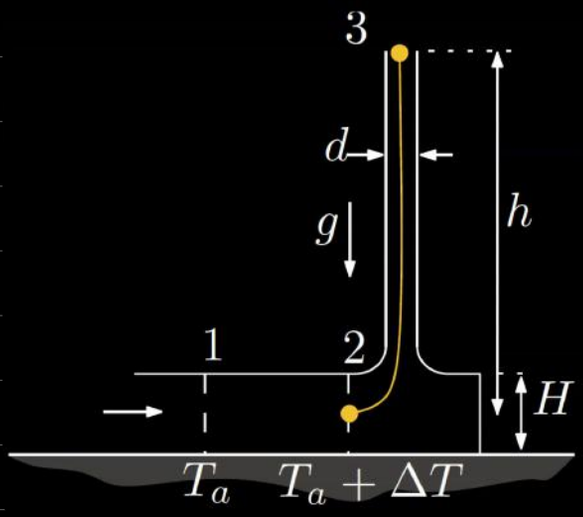
Consider a furnace of height H with a tall cylindrical smoke stack of diameter d ($d \ll H$) and height h ($h \gg H$). Air, an ideal gas ($P = \rho RT/M$), enters the furnace at atmospheric density and temperature and at local atmospheric pressure. Between stations 1 and 2, heat is added at constant pressure and the air temperature is raised by an amount ΔT. Thereafter, heat addition is negligible and the air rises through the stack at a sensibly constant density.

(a) On the assumption that viscous effects are negligible, derive an expression for the steady mass flow rate of air drawn by a stack of given height, h, in terms of the temperature rise in the furnace.

(b) If the chimney were capped off at the top, what would be the pressure differntial across the cap, assuming that ΔT would not be altered by the flow stoppage?

Note: The height h of the stack is small compared with the length RT_a/g over which the atmosphere density falls by $1/e$. Hence, gravitational density changes can be neglected.

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Solution: Let density of natural atmospheric air be ρ_a, then according to our question:

$$P_1 = P_2 = P_3 + \rho_a g h \quad (1)$$

Also Between (1) & (2):

$$\rho_a \frac{RT_a}{M} = \frac{\rho_2 R(T_a + \Delta T)}{M}$$

$$\Rightarrow \rho_2 = \rho_a T_a/(T_a + \Delta T) \quad (2)$$

Since $H << d$, $v_2 << v_3$ so for purpose of Bernoulli's equation, we may take $v_3^2 - v_2^2 \simeq v_3^2$. Now applying Bernoulli between (2) & (3):

$$P_3 + \frac{1}{2}\rho_2 v_3^2 + \rho_2 g h = P_2 \quad (3)$$

$$\Rightarrow \quad \frac{1}{2}\rho_2 v_3^2 = P_2 - P_3 - \rho_2 g h = g h(\rho_a - \rho_2) \quad (\because \text{using } (1))$$

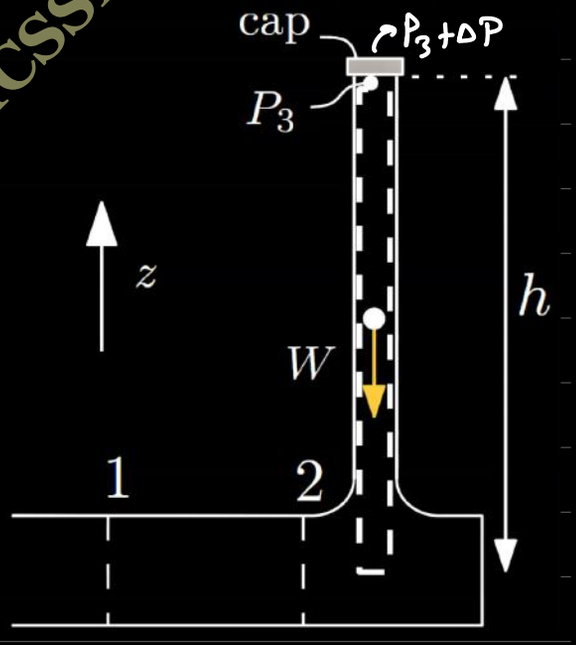
$$\Rightarrow \quad v_3 = \sqrt{2gh\left[\frac{\rho_a}{\rho_2} - 1\right]} = \sqrt{2gh\left[\frac{T_a+\Delta T}{T_a} - 1\right]} \quad (\because \text{Using } (2))$$

$$\Rightarrow \quad v_3 = \sqrt{2gh(\Delta T/T_a)}$$

Now mass flow rate $= \frac{\pi d^2}{4} \times \rho_2 \times v_3 = \frac{\pi d^2}{4} \times \rho_a\left(\frac{T_a}{T_a+\Delta T}\right)\sqrt{\frac{2gh\,\Delta T}{T_a}}$

$$\Rightarrow \quad \dot{m} = \pi d^2 \rho_a \sqrt{\frac{g h T_a \Delta T}{8(T_a+\Delta T)^2}} \quad \text{Ans. (a)}$$

(b) Now when the chimney is capped we can use normal hydrostatic pressure calculations. Calculating pressure at 1 along chimney path and free atmospheric path:



$$P_3 + \rho_2 g h = P_3 + \Delta P + \rho_a g h$$

$$\Rightarrow \quad \Delta P = (\rho_2 - \rho_a) g h$$

$$= \rho_a\left[\frac{\rho_2}{\rho_a} - 1\right] g h =$$

$$\Rightarrow \Delta P = \rho_a g h\left[\frac{T_a}{T_a+\Delta T} - 1\right] = \frac{-\rho_a g h \Delta T}{T_a+\Delta T}$$

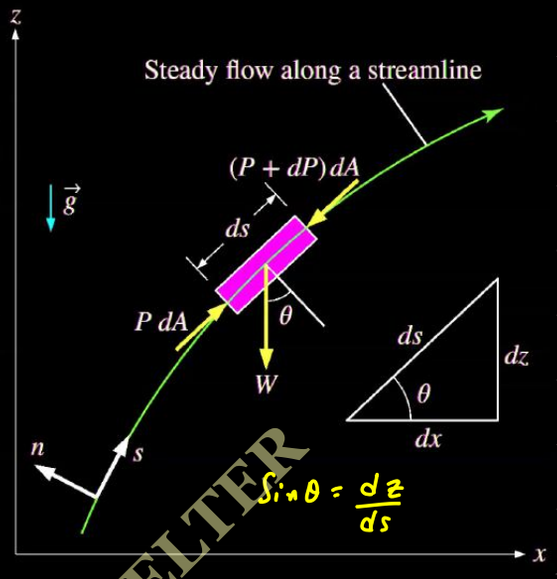
Thus pressure difference across cap: $|\Delta P| = \rho_a g h \Delta T/(T_a+\Delta T)$
(Pressure on lower side of the cap is higher than on upper side).

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Euler's Equation For Compressible Flows:

Recall that we derived Euler's equation for arbitrary steady flows as:

$$\int \frac{dP}{\rho} + \frac{v^2}{2} + g z = \text{Constant} \quad (1)$$



In case of gases we can have isothermal flows (P/ρ = constant), Isobaric flows (P = const!), Adiabatic Flows (P/ρ^{γ} = constant), Isochoric flow (ρ = constant) or general polytropic flows (P/ρ^{n} = constant). We can derive the equation for general Polytropic flow and then put in the values of n according to the nature of the flow.

Euler's Equation For Polytropic Flow

For polytropic flow, $P v^n$ = constant $\Rightarrow$ P/ρ^n = constant = C (say)

$$\Rightarrow P = C\rho^n \Rightarrow dP = Cn\rho^{n-1} d\rho \Rightarrow \frac{dP}{\rho} = Cn\rho^{n-2} d\rho$$

$$\Rightarrow Cn \int \rho^{n-2} d\rho + \frac{v^2}{2} + g z = \text{Constt.}$$

$$\Rightarrow \frac{Cn}{n-1} \rho^{n-1} + \frac{v^2}{2} + g z = \text{Constt.}$$

$$\Rightarrow \boxed{\frac{n}{n-1}\frac{P}{\rho} + \frac{v^2}{2} + g z = \text{Constt.}}$$

Alternative we could also use $\frac{P}{\rho} = \frac{RT}{M}$

$$\Rightarrow \boxed{\frac{n}{n-1}\frac{RT}{M} + \frac{v^2}{2} + g z = \text{Constt}}$$

Notice that for isochoric flows, $n = \infty$ and our equation reduces to standard Bernoulli's equation.

Our equation require $n \neq 1$, for $n = 1$ i.e isothermal flows we will have to integrate from basics.

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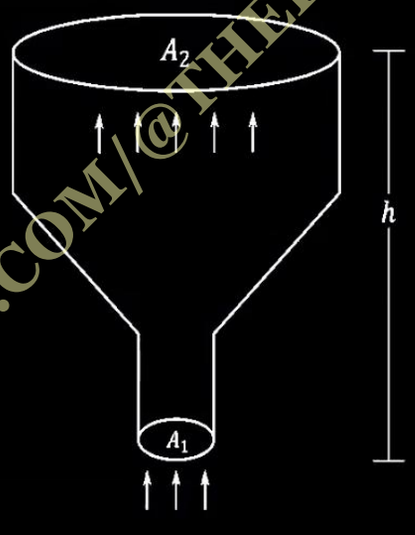
For isothermal flow $\frac{P}{\rho} = \text{const}^{tt} = C$ (say)

$\Rightarrow dP = C\,d\rho \Rightarrow \frac{dP}{\rho} = C\frac{d\rho}{\rho} \Rightarrow$ our Bernoulli's equation becomes : $C\int_{\rho_0}^{\rho}\frac{d\rho}{\rho} + \frac{v^2}{2} + gz = \text{const}^{tt} \Rightarrow C\ln\rho + \frac{v^2}{2} + gz = \text{const}^{tt}$

$\Rightarrow \frac{P}{\rho}\ln\left(\frac{\rho}{\rho_0}\right) + \frac{v^2}{2} + gz = \text{constant} \Rightarrow \frac{RT}{M}\ln\frac{\rho_2}{\rho_1} + \left(\frac{v_2^2 - v_1^2}{2}\right) + g(z_2 - z_1) = 0$

Example: JEE Advanced 2022:

An ideal gas of density $\rho = 0.2\ kg\ m^{-3}$ enters a chimney of height h at the rate of $\alpha = 0.8\ kg\ s^{-1}$ from its lower end, and escapes through the upper end as shown in the figure. The cross-sectional area of the lower end is $A_1 = 0.1\ m^2$ and the upper end is $A_2 = 0.4\ m^2$. The pressure and the temperature of the gas at the lower end are 600 Pa and 300 K, respectively, while its temperature at the upper end is 150 K. The chimney is heat insulated so that the gas undergoes adiabatic expansion. Take $g = 10\ m\ s^{-2}$ and the ratio of specific heats of the gas $\gamma = 2$. Ignore atmospheric pressure.

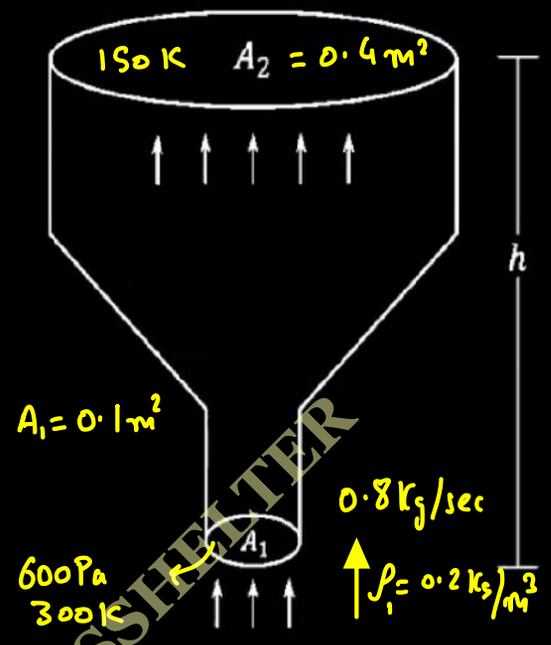


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Which of the following statement(s) is(are) correct?

(A) The pressure of the gas at the upper end of the chimney is 300 Pa.
(B) The velocity of the gas at the lower end of the chimney is 40 $m\ s^{-1}$ and at the upper end is 20 $m\ s^{-1}$.
(C) The height of the chimney is 590 m.
(D) The density of the gas at the upper end is 0.05 $kg\ m^{-3}$.

Solution: As clear from the language of question, the situation here is that of adiabatic steady flow i.e $n = \gamma = 2$.



Now for adiabatic flow, $P^{1-n} T^{n} = \text{Constt.}$

Thus for points ① & ②:

$$600^{1-2} \times 300^{2} = P_2^{1-2} \times 150^{2}$$

$$\Rightarrow P_2 = \left(\frac{150}{300}\right)^2 \times 600 = 150\ Pa$$

Thus A is incorrect.

Now $\dot{m} = \rho_1 A_1 v_1 = \rho_2 A_2 v_2$

$$\Rightarrow v_2 = \frac{\rho_1}{\rho_2}\frac{A_1}{A_2} v_1 \quad \text{Now } P/\rho^{\gamma} = \text{Constt} \Rightarrow P/\rho^{2} = \text{Constt}$$

$$\Rightarrow \frac{\rho_1}{\rho_2} = \sqrt{\frac{P_1}{P_2}} = \sqrt{\frac{600}{150}} = 2$$

Now $v_1 = \dot{m}/\rho_1 A_1 = 0.8/(0.2 \times 0.1) = 40\ ms^{-1}$

$$\Rightarrow v_2 = 2 \times \frac{1}{4} \times 40 = 20\ ms^{-1}$$ Thus option B is correct.

Now Euler equation for steady flow:

$$\frac{n}{n-1}\frac{P}{\rho} + \frac{v^2}{2} + gz = \text{Constt.}$$

Now $\dot{m} = \rho_2 A_2 v_2 \Rightarrow \rho_2 = \dot{m}/A_2 v_2 = 0.8/(0.4 \times 20) = 0.1\ Kg/m^3$

Thus Between ① & ②:

$$2 \times \frac{600\ Pa}{0.2\ Kg/m^3} + \frac{40^2}{2} + 10 \times 0 = 2 \times \frac{150}{0.1} + \frac{20^2}{2} + 10 \times h$$

$$\Rightarrow h = [6000 + 400 - 3000 - 200]/10 = 360\ m$$

Thus option C is incorrect, D is also incorrect ($\rho_2 = 0.1\ Kg/m^3$).

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Example: IroDov 2.42:

Find the rate v with which helium flows out of a thermally insulated vessel into vacuum through a small hole. The flow rate of the gas inside the vessel is assumed to be negligible under these conditions. The temperature of helium in the vessel is $T = 1{,}000$ K.



Solution: Consider the volume of Helium in the nozzle. This volume is being pushed out due to pressure inside the tank. The process is adiabatic because the tank is insulated. Now applying Euler's equation for adiabatic flow:

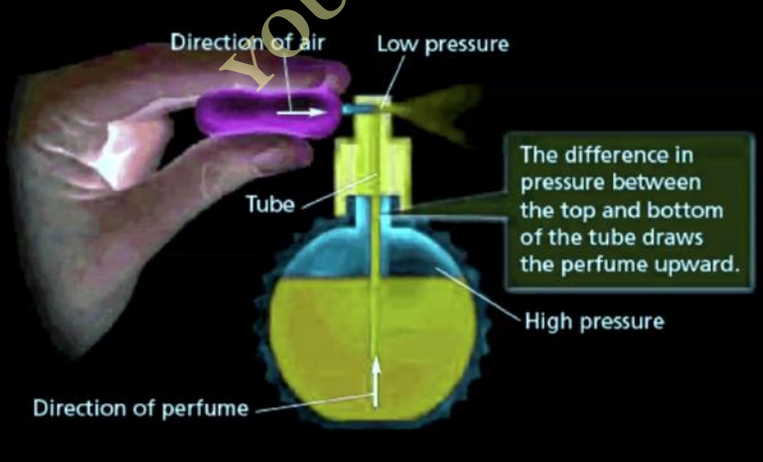
$$\frac{\gamma}{\gamma-1}\frac{P}{\rho} + \frac{v^2}{2} + gz = \text{Constant}.$$

For our case, z is irrelevant (∵ gas and small heights) velocity inside the container is nearly zero. In vacuum, pressure is zero. $\gamma = 1 + 2/3 = 5/3$

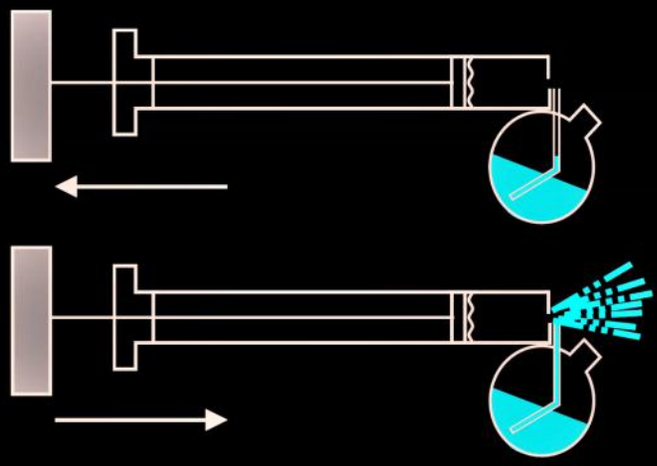
$\Rightarrow \gamma/\gamma-1 = (5/3)/(2/3) = 5/2$, also $P/\rho = RT/M$. Thus:

$$\frac{5}{2}\frac{RT}{M} + 0 = 0 + \frac{v^2}{2} \Rightarrow v = \sqrt{5RT/M} \quad \text{i.e.} \quad \sqrt{\frac{2\gamma}{\gamma-1}\frac{RT}{M}} \quad \text{Ans.}$$

Atomizer / Spray Gun:



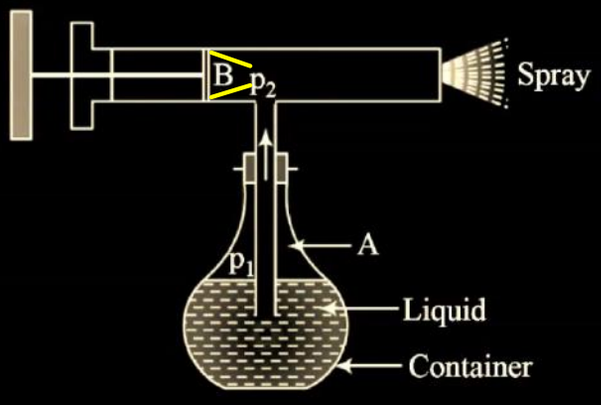
A



B

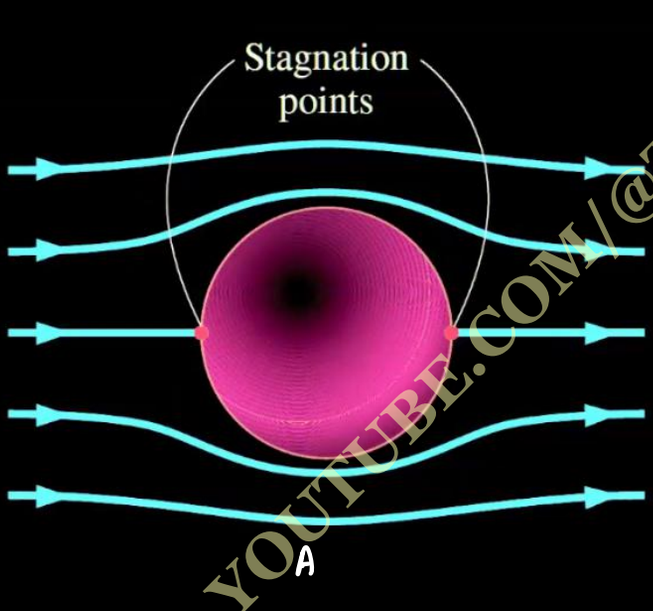
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The figures A, B and C show various schematics of atomisers / spray guns. In all of these mechanisms, low pressure is achieved near mouth of a tube containing the liquid which is to be sprayed. Recall that a consequence of Bernoulli's equation is that high velocity leads to a low pressure. The high velocity is achieved using tubes where cross section reduces near the location where we wish low pressure to be achieved. The low pressure creates a suction force because of which liquid mixes with air stream and get sprayed.

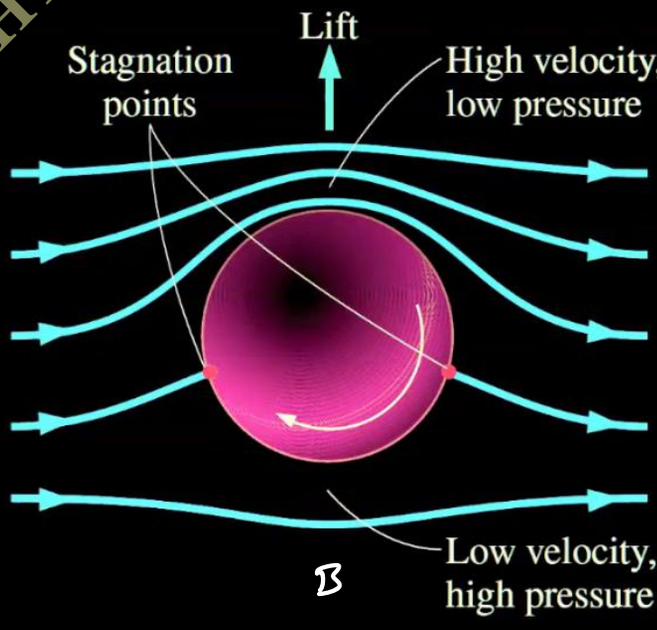


C

Lift Force On A Spinning Ball : (Magnus Effect)



A



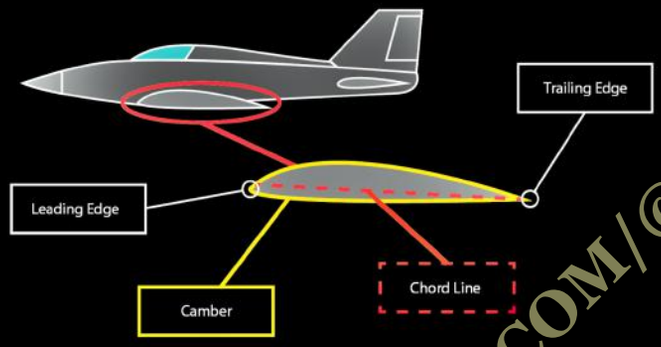
B

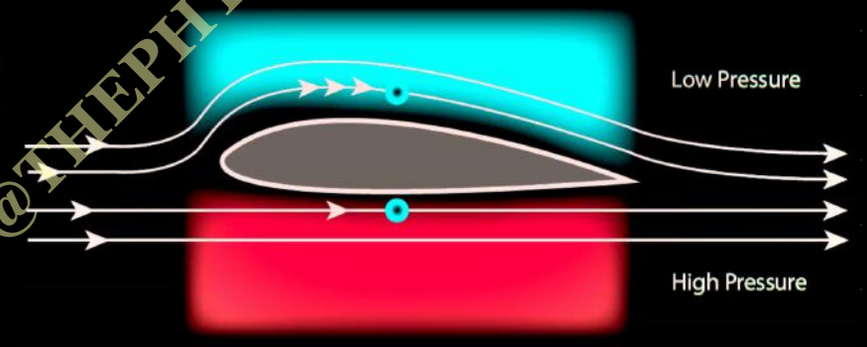
Consider a cricket ball moving with certain instantaneous velocity leftwards. The above figures show streamlines from a translating frame moving with CM of the ball for two cases : (A) Without any spin and (B) with clockwise spin. Due to roughness of the ball, the surface of the ball speeds up the air at the tops while slows down the air below its bottom. Thus there is crowding of streamlines at the top leading to a lower pressure.

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similarly below the ball there will be a higher pressure, the net result of these effects is that there is a net upward force on the ball leading to a flighted trajectory, i.e it travels a greater distance before falling to the ground. By suitably changing the direction of angular velocity vector we can also generate downward push or sideways deflections also.
This swinging of ball is known as magnus effect. The swing of the ball can also be generated by polishing only one side of the ball, the polished side does not have as much effect on wind velocity as the rough side. (That's why you sometimes fielders polishing one side of the ball before throwing it back to the bowler).

AEROFOIL EFFECT:





The shape of streamlines around wings of an aircraft from the aircraft frame is shown in the figure above. Because of shape of the wing, the streamlines above the wing are more crowded thus there is higher velocity on the upper side of the wing than on lower side. This creates a low pressure area above the wing as compared to area render it thus creating a net lift force on the aircraft.

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